

The Importance of Inventory Management — Complete Solution Report

Lecture 01 · Concepts, input data, calculations and visual analysis for every topic of the lecture

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1. U.S. logistics costs and the weight of inventory (2000–2014)

The U.S. economy held between **\$1.4 and \$2.5 trillion** of business inventory each year in 2000–2014. Holding it cost **19–25 % of its value per year** (the *inventory carrying rate*), making inventory carrying cost roughly one-third of all U.S. logistics spending.

$c = a \times b$	$f = c + d + e$	share of GDP = f / GDP
\$2,496 bn Inventory held in 2014	\$477 bn Carrying cost in 2014	34.0 % Inventory share of logistics cost (avg)
\$1,019 → \$1,433 bn Total logistics cost 2000 → 2014	7.8 % (2009) Lowest logistics share of GDP	3.58 % / yr Inventory value CAGR 2000–2014
0.49 bn (rounding) Max check error $c = a \times b$	0.00 bn Max check error $f = c + d + e$	

U.S. logistics cost composition, 2000–2014

Transportation is the largest component; inventory carrying is a steady ~1/3 of the total

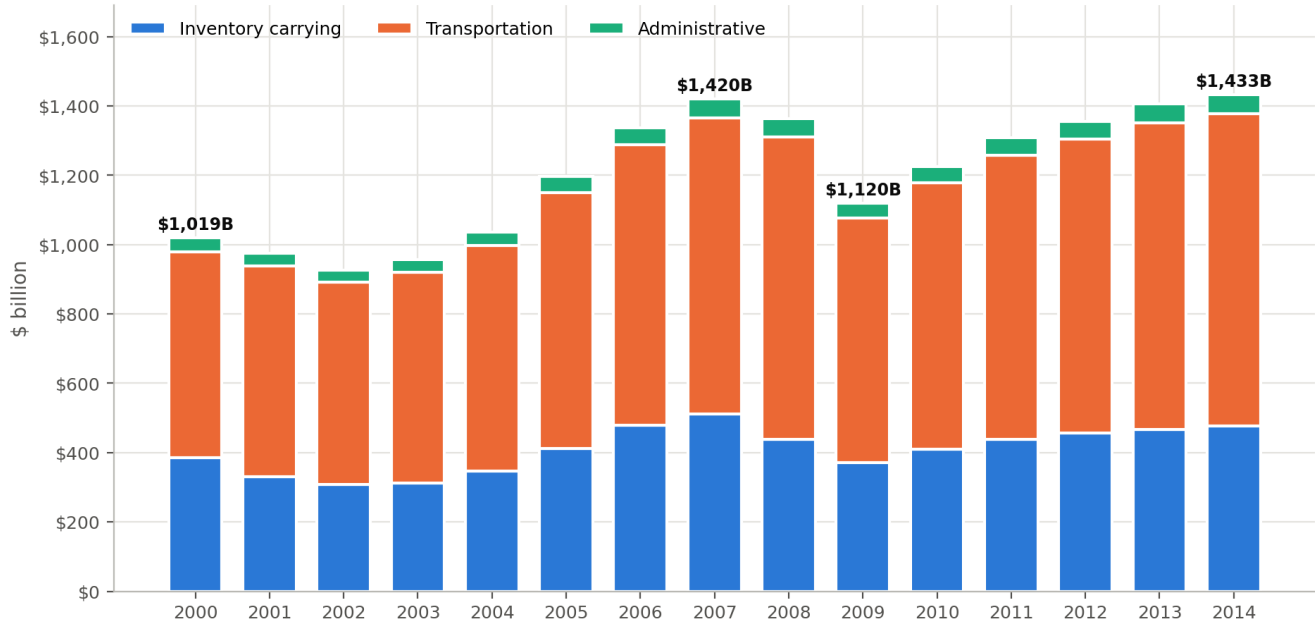


Figure 1.1 — Composition of U.S. logistics costs (\$ billion). Source: lecture slide 5.

Inventory indicators move with the business cycle

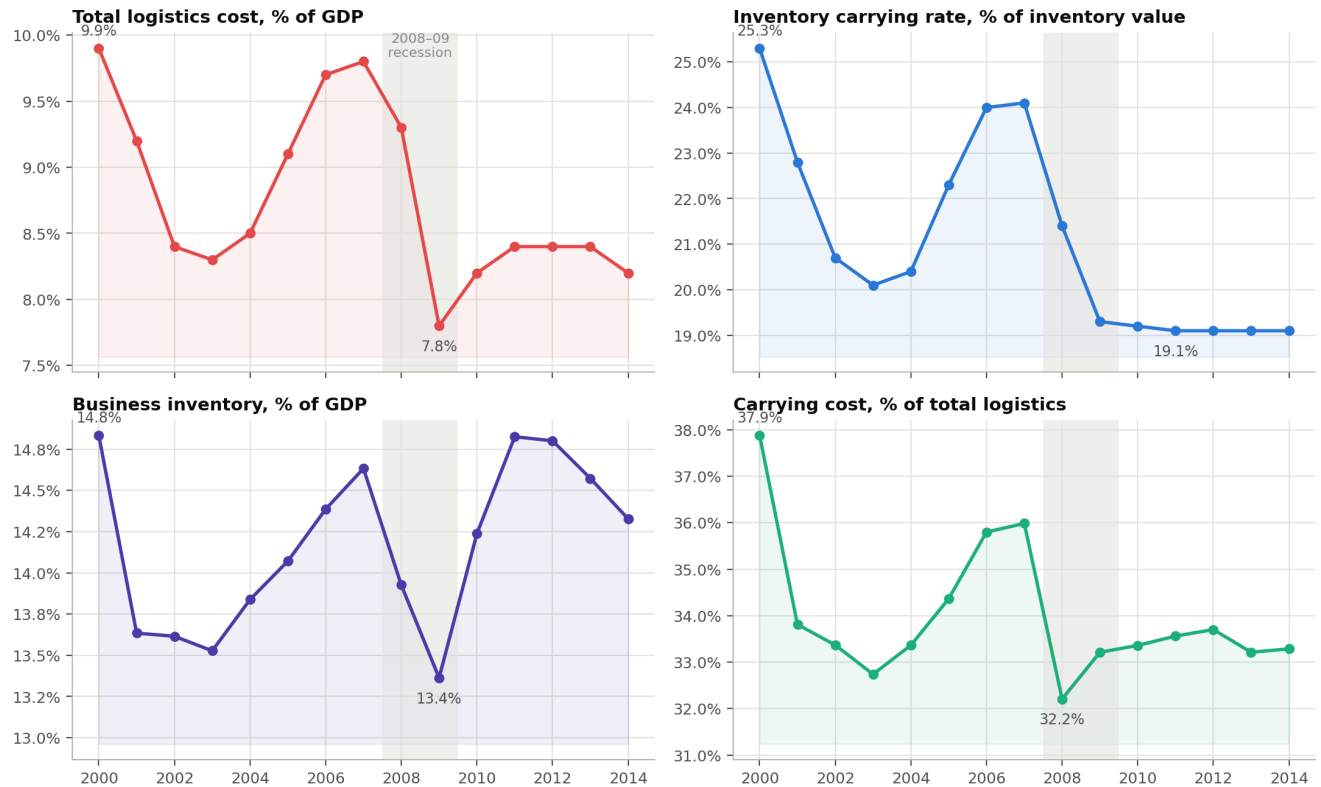


Figure 1.2 — Key ratios derived from the lecture data. The 2008–09 recession (shaded) cut both inventory and logistics intensity.

Growth indexed to 2000

Inventory value grew ~64 % while its carrying cost grew only ~24 %: cheaper capital lowered the carrying rate

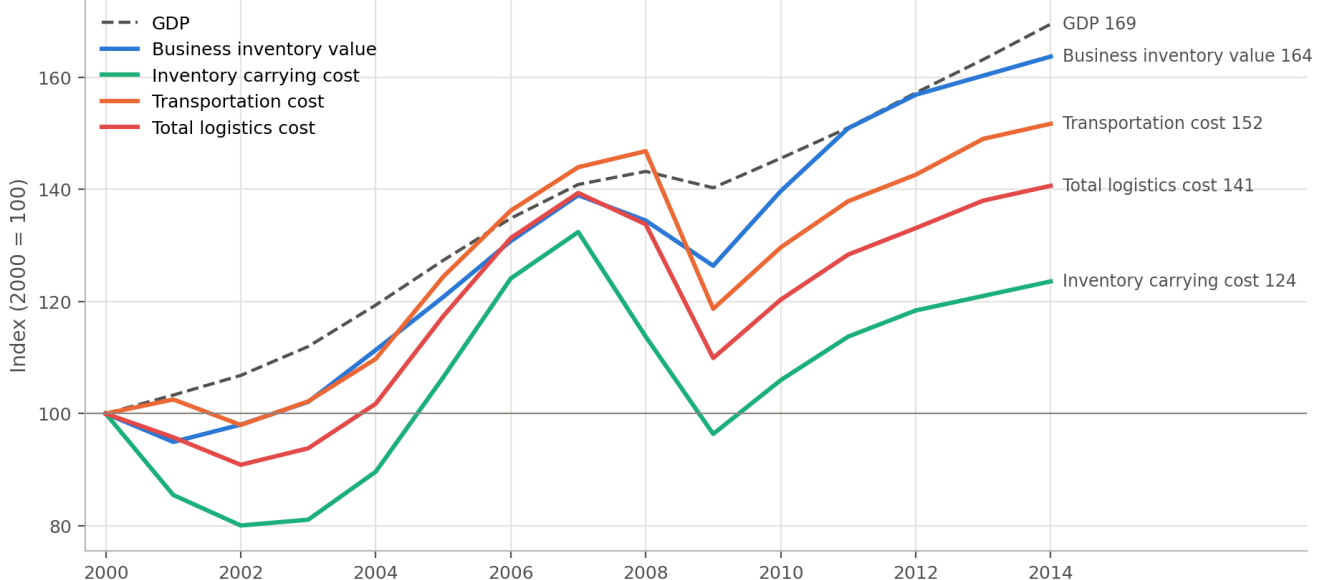


Figure 1.3 — All series indexed to 2000 = 100 so that different magnitudes share one axis.

Lecture data with verification columns ($\text{check_carrying_bn} = a \times b$) and derived shares.

year	gdp bn	business inventory bn	inventory carrying rate pct	inventory carrying cost bn	check carrying bn	transportation cost bn	administrative cost bn	total logistics cost bn	logistics pct gdp	inv share of logistics pct
2000	10,280	1,525	25.3	386	385.8	594	39	1,019	9.9	37.9
2001	10,620	1,448	22.8	330	330.1	609	37	976	9.2	33.8
2002	10,980	1,495	20.7	309	309.5	582	35	926	8.4	33.4
2003	11,510	1,557	20.1	313	313.0	607	36	956	8.3	32.7
2004	12,270	1,698	20.4	346	346.4	652	39	1,037	8.5	33.4
2005	13,090	1,842	22.3	411	410.8	739	46	1,196	9.1	34.4
2006	13,860	1,994	24.0	479	478.6	809	50	1,338	9.7	35.8
2007	14,480	2,119	24.1	511	510.7	855	54	1,420	9.8	36.0
2008	14,720	2,050	21.4	439	438.7	872	52	1,363	9.3	32.2
2009	14,420	1,927	19.3	372	371.9	705	43	1,120	7.8	33.2
2010	14,960	2,130	19.2	409	409.0	770	47	1,226	8.2	33.4
2011	15,520	2,301	19.1	439	439.5	819	50	1,308	8.4	33.6
2012	16,160	2,392	19.1	457	456.9	847	52	1,356	8.4	33.7
2013	16,770	2,444	19.1	467	466.8	885	54	1,406	8.4	33.2
2014	17,420	2,496	19.1	477	476.7	901	55	1,433	8.2	33.3

Takeaway. Inventory is not just a warehouse issue — at a national scale it represents hundreds of billions of dollars of cost each year. Even a one-point reduction in the carrying rate on \$2.5 trillion of inventory saves about **\$25 billion per year**.

2. Inventory-to-sales ratios and the downstream shift of inventory

The I/S ratio converts inventory into *months of sales*. A falling ratio means the same sales are supported with less stock (leaner operations); a spike signals that sales dropped faster than production could be cut.

$$\text{I/S ratio} = \text{Inventory (\$)} / \text{Monthly sales (\$)}$$

1.55 → 1.09

Manufacturing I/S 2016 → 2025

65.9 % → 74.7 %

Downstream share of inventory

1.32 → 1.35

Wholesale I/S 2016 → 2025

1.84 (Aug 2020)

Peak I/S (any echelon)

1.62 → 1.76

Retail I/S 2016 → 2025

120

Months analysed

Inventory-to-sales ratio by echelon

Manufacturing grows leaner while downstream echelons carry relatively more stock

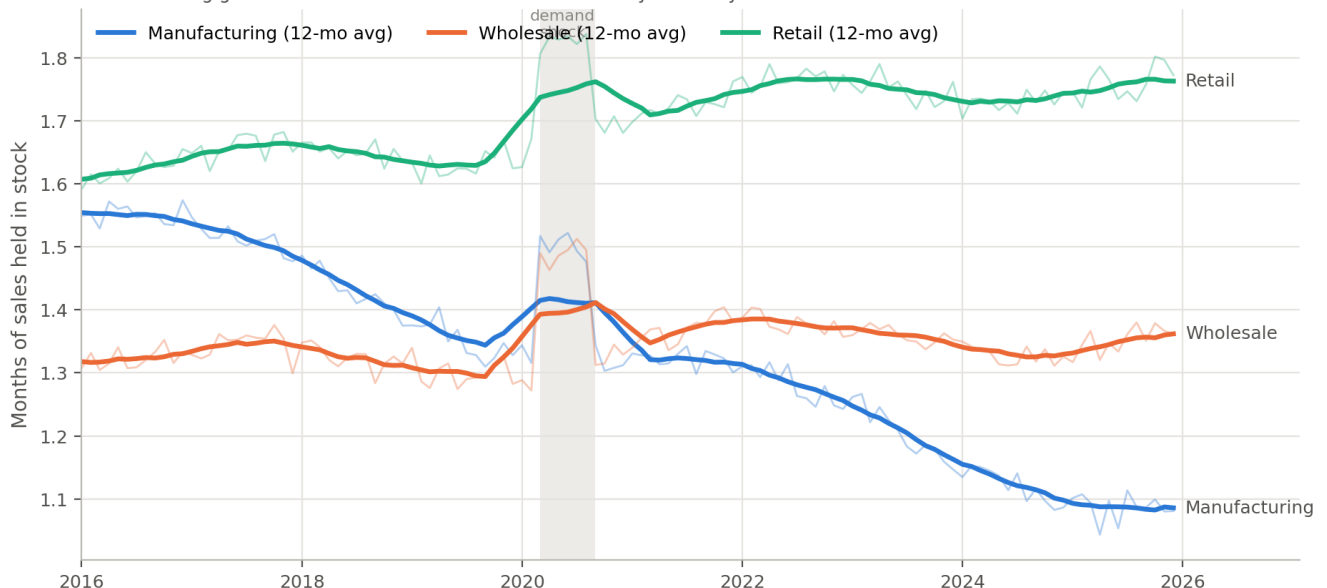


Figure 2.1 — Monthly I/S ratio (thin) and 12-month centred moving average (bold).

Where is the inventory held?

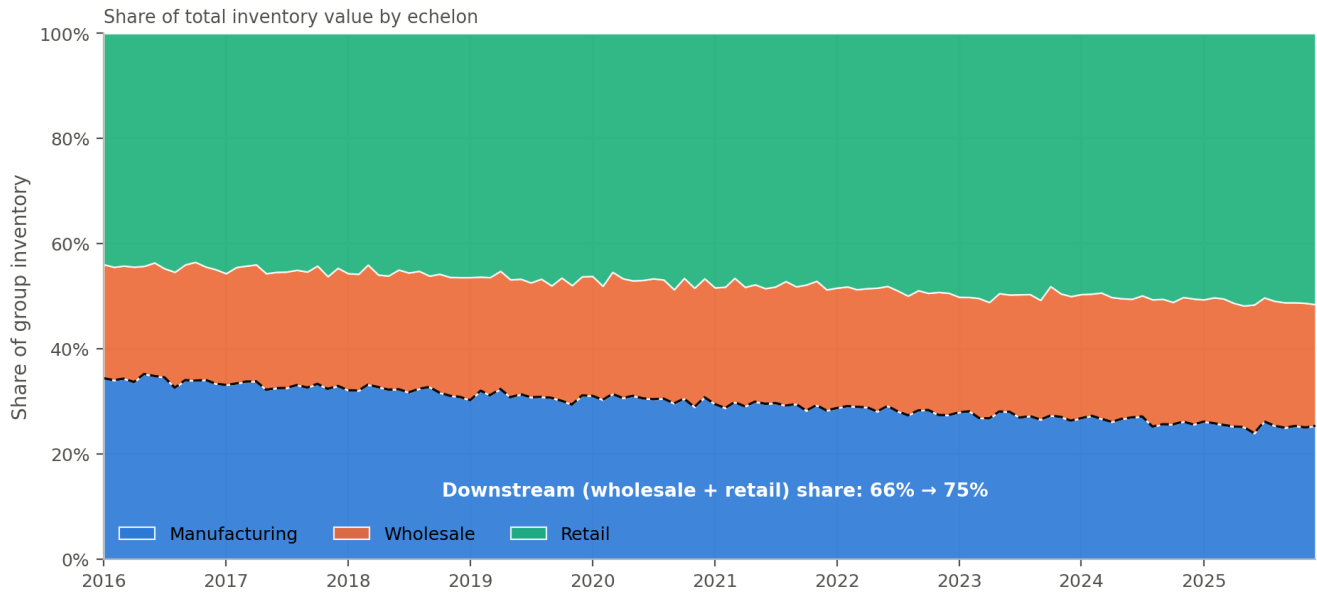


Figure 2.2 — Fraction of total inventory held at each echelon; dashed line = boundary between manufacturing and downstream stock.

Annual inventory-to-sales ratio heat map

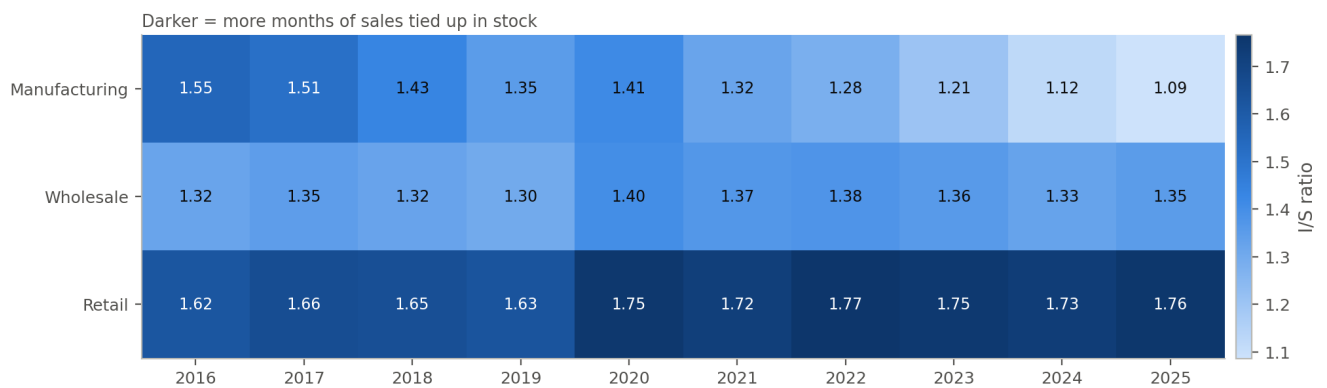


Figure 2.3 — Annual I/S ratio = $\Sigma \text{inventory} / \Sigma \text{sales within the year}$.

Annual I/S ratios and downstream share of inventory.

Year	Manufacturing	Wholesale	Retail	Downstream share %
2016	1.552	1.322	1.621	65.9
2017	1.512	1.345	1.659	67.0
2018	1.432	1.325	1.651	67.9
2019	1.352	1.299	1.629	69.1
2020	1.414	1.402	1.755	69.5
2021	1.323	1.369	1.723	70.8
2022	1.282	1.378	1.766	71.7
2023	1.205	1.360	1.750	72.8
2024	1.122	1.328	1.730	73.7
2025	1.087	1.352	1.760	74.7

- **Global sourcing** lengthens replenishment lead times, so wholesalers and retailers must hold more pipeline and safety stock.
- **Product variety** multiplies SKUs at the point of sale; each SKU needs its own safety stock (see Part 3.2).
- Manufacturers adopting lean / make-to-order practices push finished-goods stock downstream.

3. The business cycle and aggregate inventory investment

Parameters: cycle length **24 months**, amplitude **±15 %**, production leads sales by **3 months**, starting inventory **150 units**, contribution margin **25 %**, holding cost **2 % per month**.

$$I(t) = I_0 + \sum [P(k) - S(k)]$$

183

Max inventory

95

Min inventory

88 units = 88% of a month's sales

Swing (max – min)

11.4

Largest monthly build-up

-11.4

Largest monthly disinvestment

22

Inventory peaks ≈ months after sales peak

Production leads sales → inventory builds before the peak and is drawn down after it

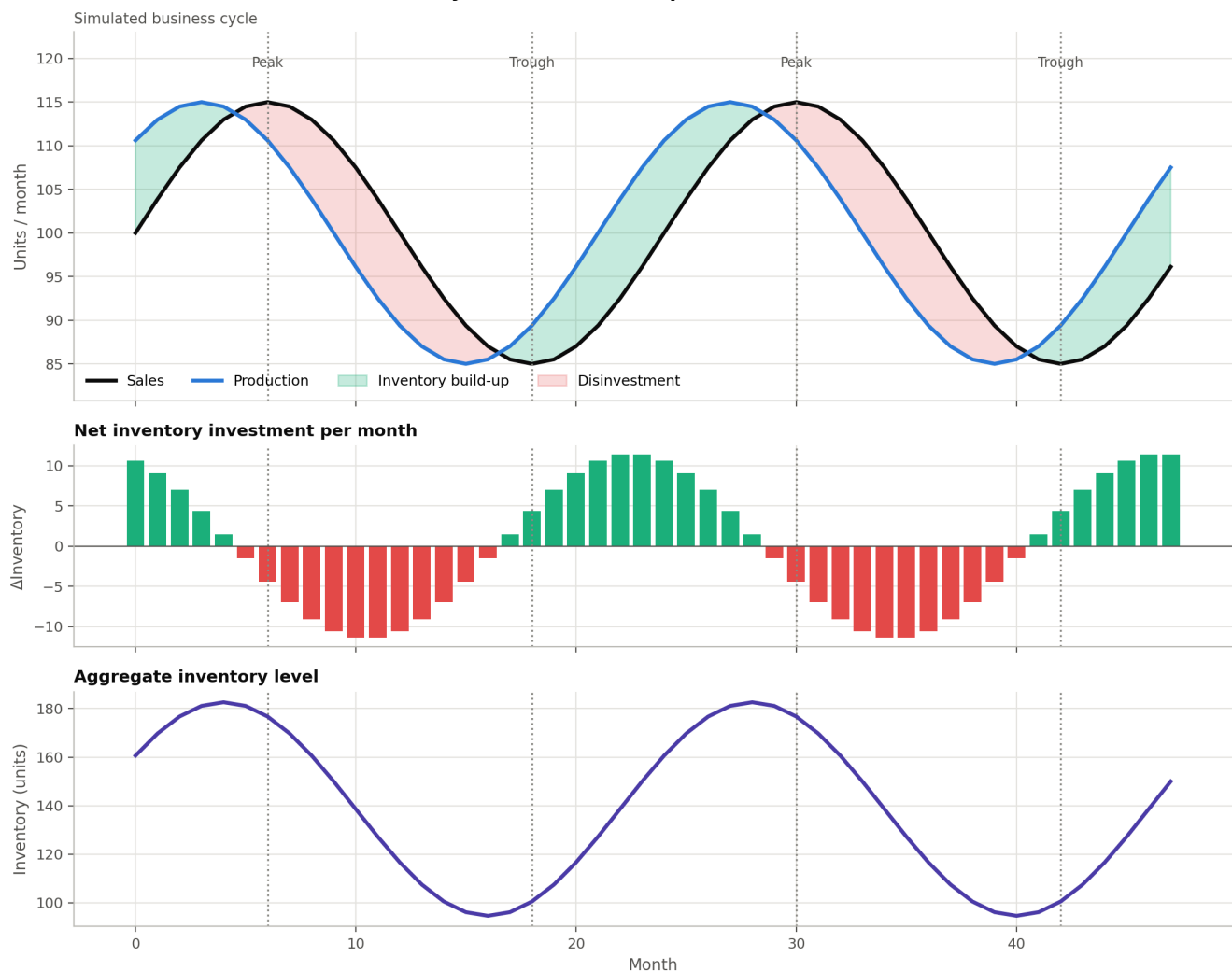


Figure 3.1 — Business-cycle simulation: sales vs production, monthly inventory investment and inventory level.

Corporate profits

Profits fall after the peak: lower sales plus the cost of carrying the excess stock

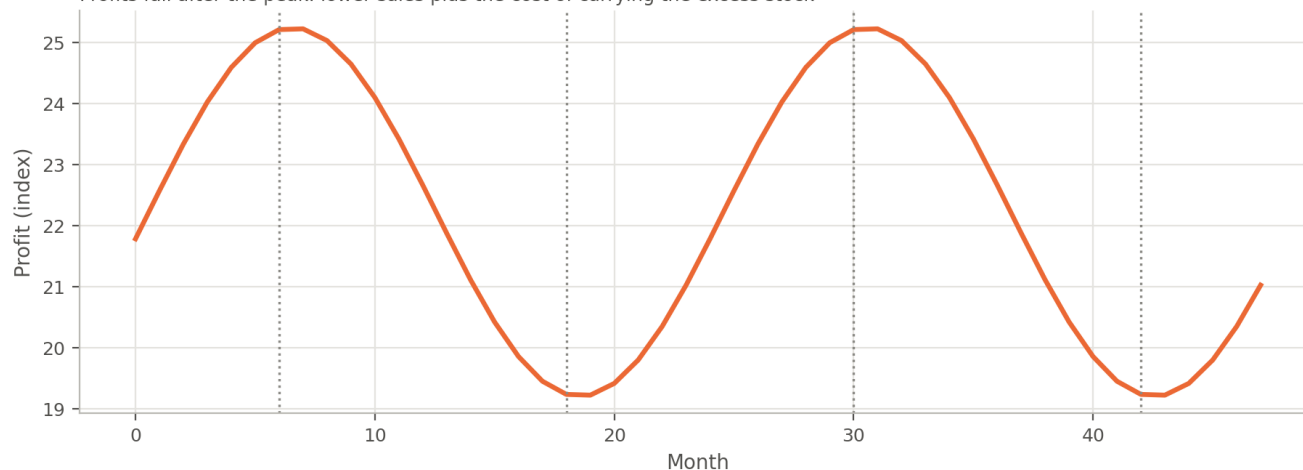


Figure 3.2 — Profit = margin × sales – holding cost × inventory.

Simulation output (every 3rd month).

month	sales	production	inventory change	inventory	profit
1	100.0	110.6	+10.6	160.6	21.79
4	110.6	115.0	+4.4	181.1	24.03
7	115.0	110.6	-4.4	176.7	25.22
10	110.6	100.0	-10.6	150.0	24.65
13	100.0	89.4	-10.6	116.6	22.67
16	89.4	85.0	-4.4	96.1	20.43
19	85.0	89.4	+4.4	100.5	19.24
22	89.4	100.0	+10.6	127.2	19.80
25	100.0	110.6	+10.6	160.6	21.79
28	110.6	115.0	+4.4	181.1	24.03
31	115.0	110.6	-4.4	176.7	25.22
34	110.6	100.0	-10.6	150.0	24.65
37	100.0	89.4	-10.6	116.6	22.67
40	89.4	85.0	-4.4	96.1	20.43
43	85.0	89.4	+4.4	100.5	19.24
46	89.4	100.0	+10.6	127.2	19.80

4. Cycle stock: batch size, cycle frequency and average inventory

Annual demand $D = 12,000$ units over 240 working days $\rightarrow$ demand rate $d = 50.0$ units/day.

$$\text{Average cycle stock} = Q / 2 = D / (2n)$$

4,000 units $\rightarrow$ avg 2,000

Largest lot (Q)

50.0 units/day

Demand rate

3 $\rightarrow$ 6 cycles

Doubling cycles/year

50 %

Avg cycle stock falls by

Saw-tooth cycle stock: smaller, more frequent lots cut average inventory

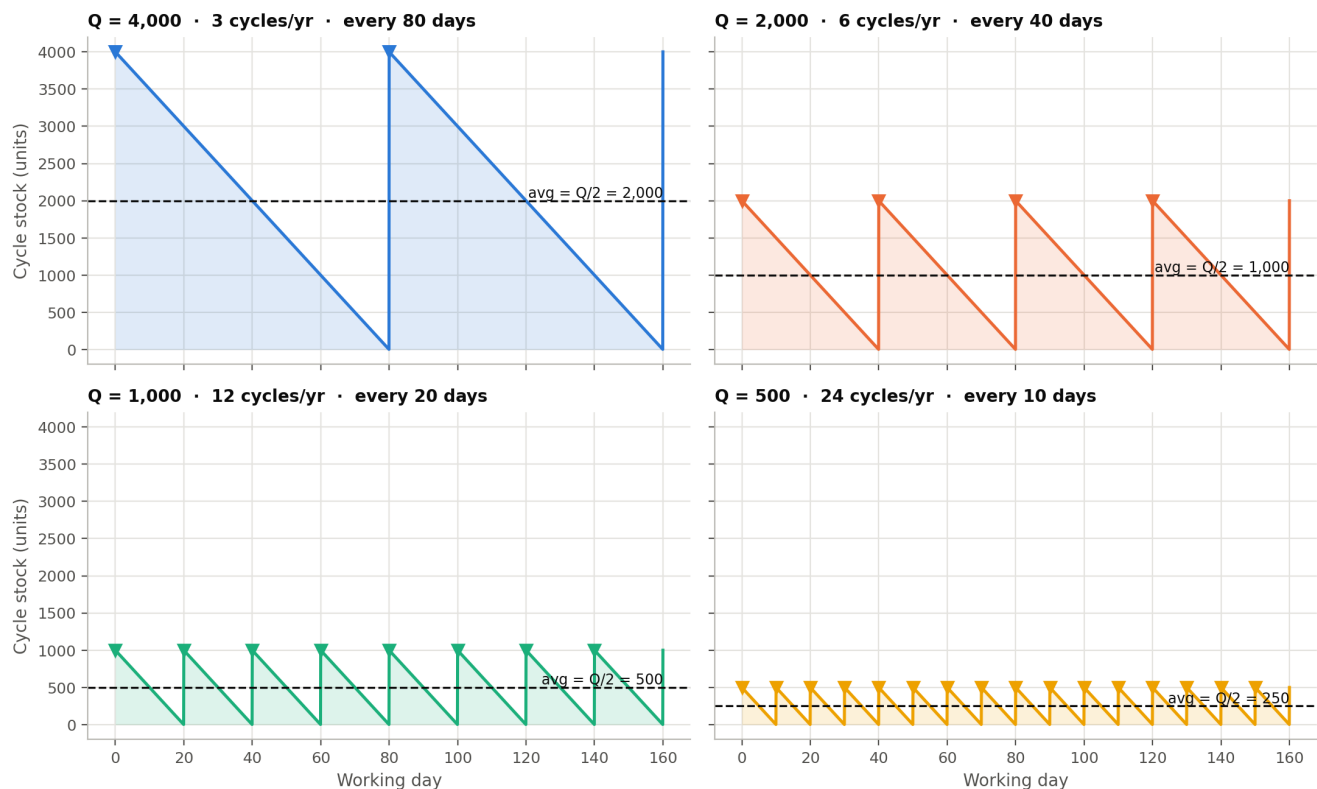


Figure 4.1 — Cycle-stock profiles for the four lot sizes. ▼ marks a make/order event; dashed line = average ($Q/2$).

Doubling the number of cycles halves average cycle stock (fixed base stock unchanged)

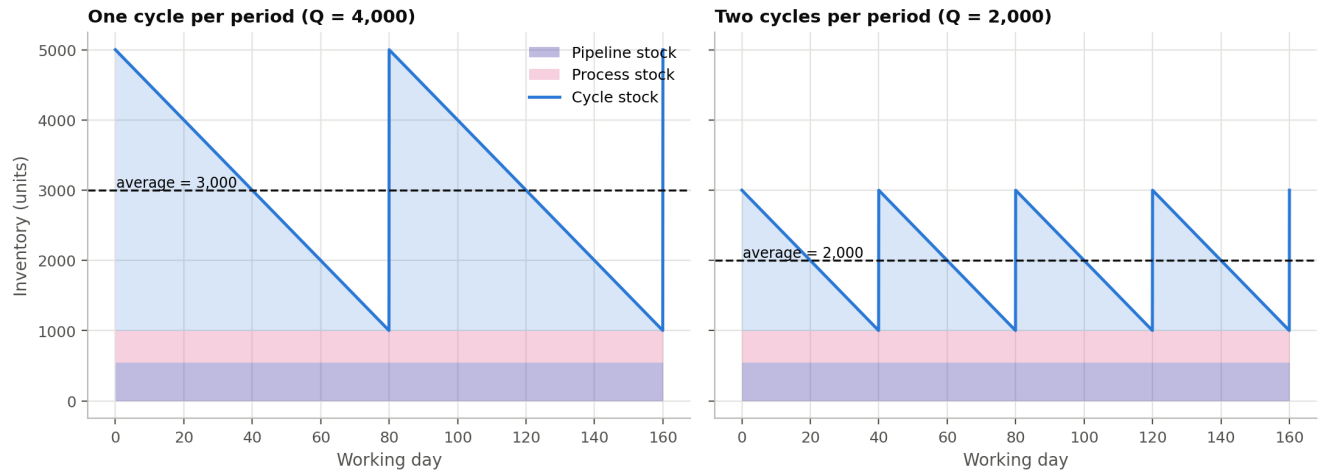


Figure 4.2 — Replica of lecture slide 19 with pipeline and process stock as a fixed base.

Average cycle stock vs cycle frequency

$\bar{I} = D / 2n$ — a hyperbola: the first extra cycles give the biggest reduction, but each extra cycle costs a set-up / order

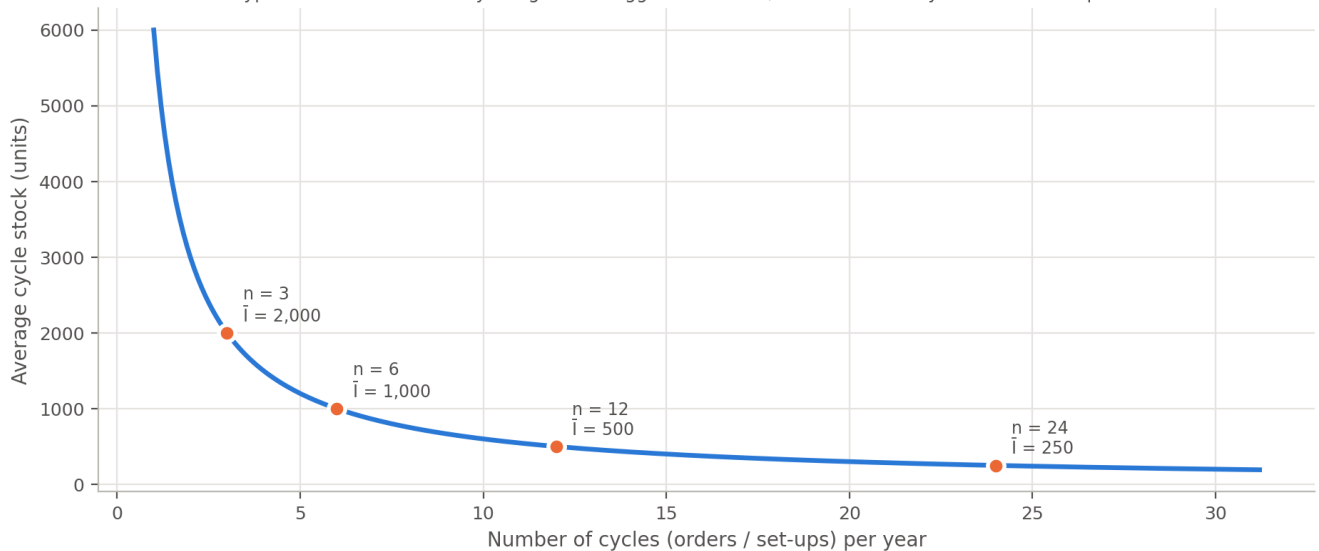


Figure 4.3 — Trade-off preview: fewer units in stock requires more frequent set-ups (see Part 6.4).

Cycle stock for alternative lot sizes.

lot size Q	cycles per year	cycle time days	max cycle stock	min cycle stock	avg cycle stock units	avg coverage days
4,000	3.0	80.0	4,000	0	2,000	40.0
2,000	6.0	40.0	2,000	0	1,000	20.0
1,000	12.0	20.0	1,000	0	500	10.0
500	24.0	10.0	500	0	250	5.0

5. Safety stock: forecast error, supply error and service level

Mean daily demand **198.9**, forecast-error σ_d **45.9** units/day; mean lead time **7.75 days** (quoted 8 days), σ_L **1.77** days.

$$\sigma_{DLT} = \sqrt{L \cdot \sigma_d^2 + d^2 \cdot \sigma_L^2}$$

$$SS = z \cdot \sigma_{DLT}$$

$$ROP = d \cdot L + SS$$

128 units

σ of lead-time demand (demand error only)

2,158 units

Reorder point for 95 % CSL

193 % to SS

Supply variability adds

375 units

σ of lead-time demand (demand + supply)

93.3 %

Simulated CSL at that ROP

617 units

SS for 95 % CSL

0 units

SS with certain demand and lead time

Why safety stock is needed: demand never equals the forecast

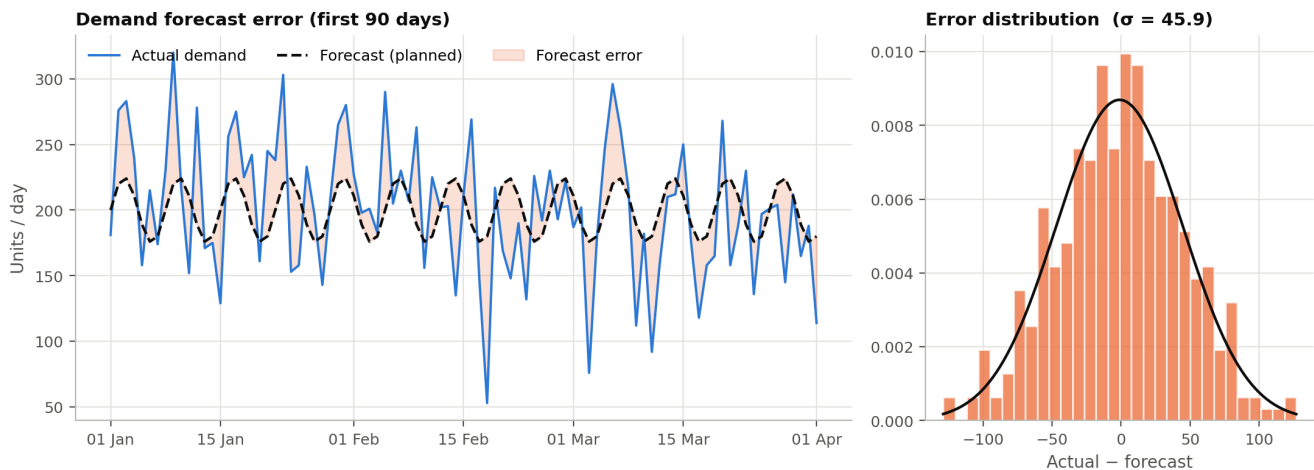


Figure 5.1 — Daily forecast vs actual demand and the histogram of forecast errors with a fitted normal curve.

Supply forecast error: actual vs quoted lead time

35 % of orders arrived later than quoted — timing uncertainty also needs protection

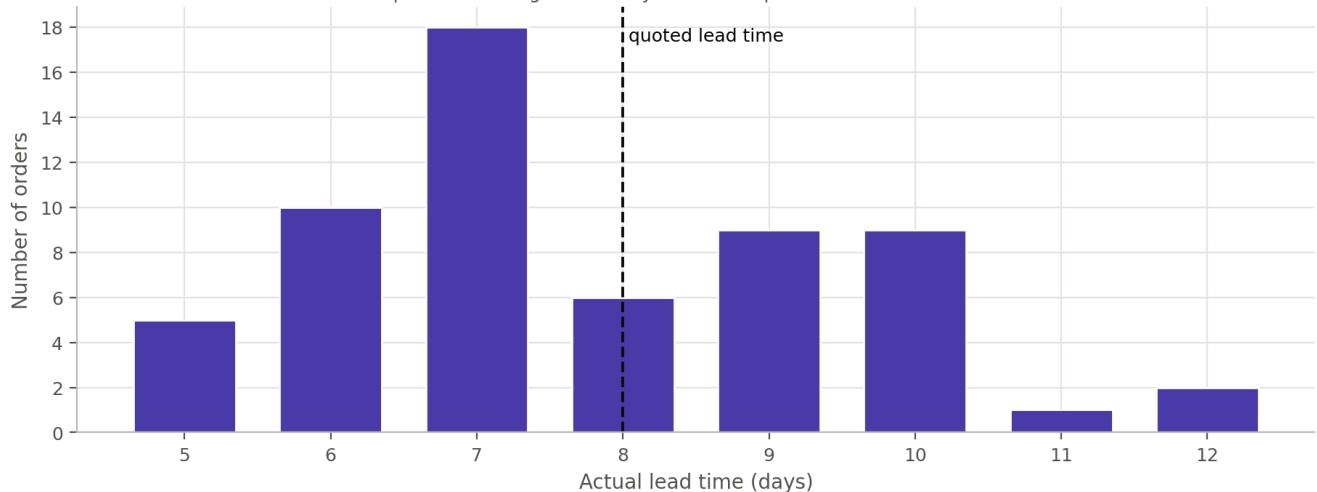


Figure 5.2 — Distribution of 60 observed replenishment lead times.

Demand during the lead time and the reorder point

Safety stock = distance between the expected lead-time demand and the reorder point

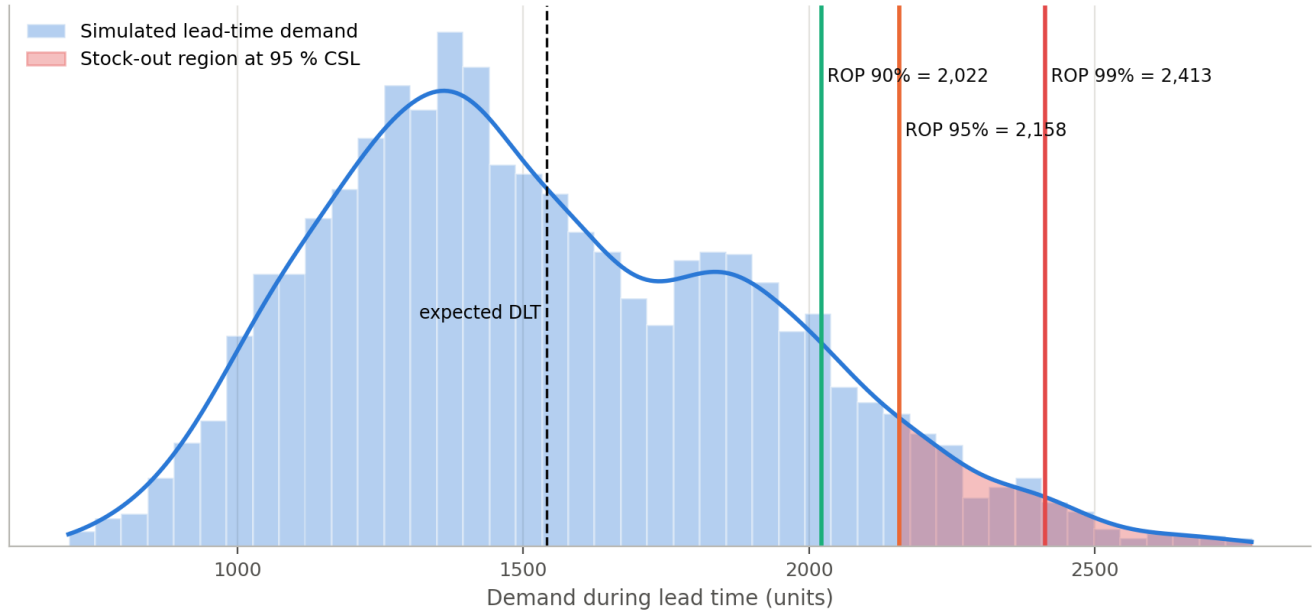


Figure 5.3 — Monte-Carlo distribution of lead-time demand (4,000 bootstrap replications).

Safety stock is directly related to the desired customer service level

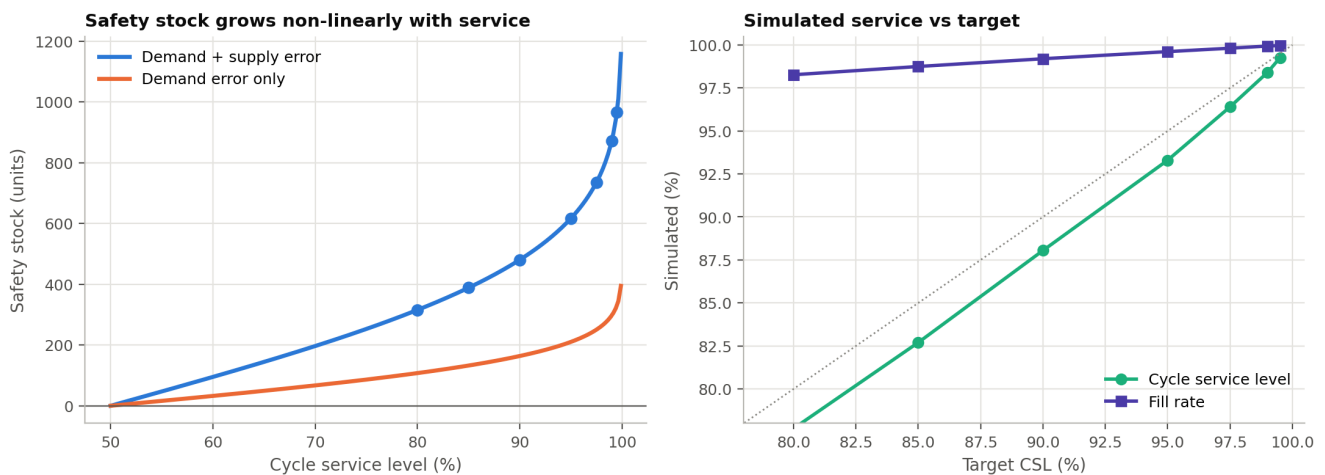


Figure 5.4 — Left: $SS = z \cdot \sigma$. Right: simulated cycle service level and fill rate (order quantity $Q = 2,983$ units).

Same demand, same lead times — safety stock absorbs the variability

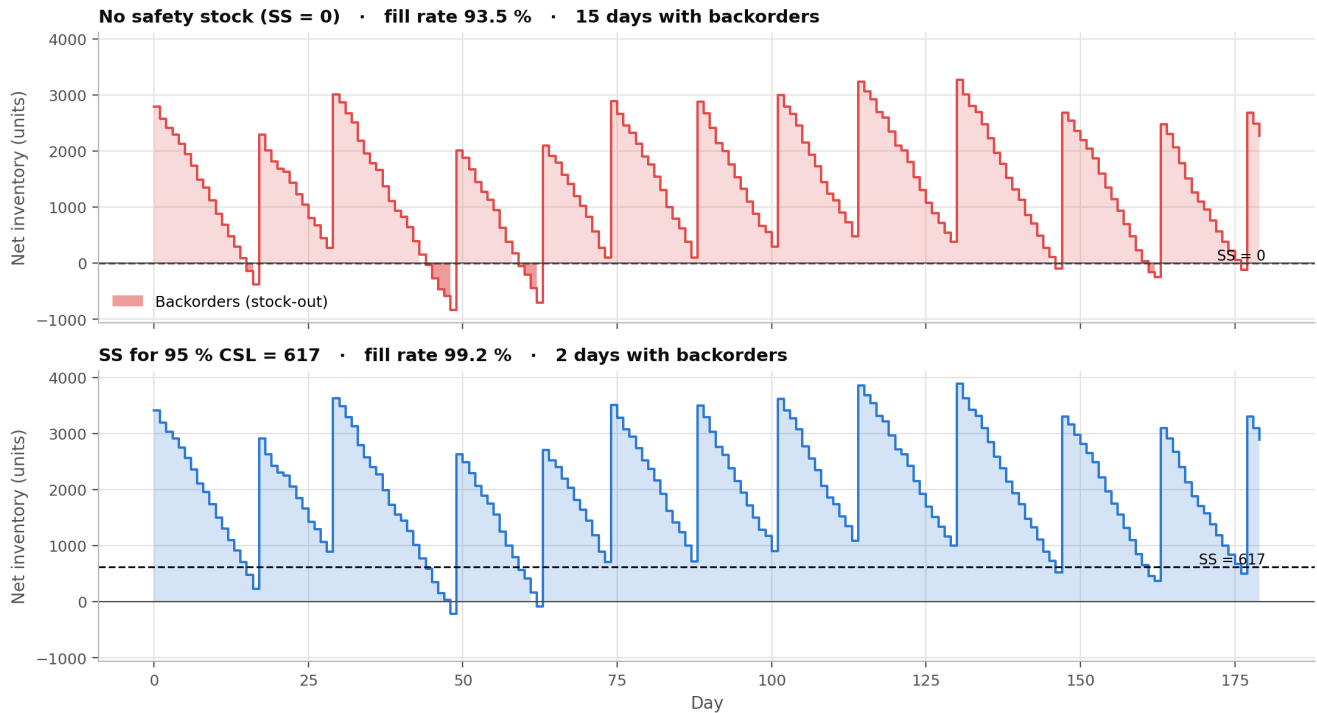


Figure 5.5 — 180-day simulation of a continuous-review (s, Q) policy with bootstrapped demand and lead times.

Reading the simulation. The analytic SS assumes lead-time demand is normal. The simulated (bootstrap) lead-time demand is skewed by the discrete, right-tailed lead times, so the achieved CSL is a little below target at mid service levels — a reminder to validate formula-based safety stocks against real data. The fill rate is always higher than the CSL because a stock-out cycle usually misses only part of the demand.

Safety stock, reorder point and simulated service by target cycle service level.

cycle service level	z	SS demand error only	SS demand and supply error	ROP	annual SS holding cost	simulated CSL	simulated fill rate
80.0%	0.842	108	315	1,857	6,420	77.7%	98.26%
85.0%	1.036	132	388	1,930	7,906	82.7%	98.75%
90.0%	1.282	164	480	2,022	9,775	88.0%	99.20%
95.0%	1.645	210	617	2,158	12,547	93.3%	99.61%
97.5%	1.960	250	735	2,276	14,950	96.4%	99.81%
99.0%	2.326	297	872	2,413	17,745	98.4%	99.93%
99.5%	2.576	329	966	2,507	19,648	99.2%	99.97%

Simulation comparison.

policy	reorder point	fill rate pct	days with backorders	avg on hand
No safety stock (SS = 0)	1,541	93.5	15	1,387
SS for 95 % CSL = 617	2,158	99.2	2	1,979

6. Total stock: cycle, safety, process, pipeline and anticipation stock

Pipeline stock = daily demand × transit days (Little's law); Cycle = $\frac{1}{2} \times$ cycle days

\$331,095

Total inventory value

23.1 %

Safety share of value

P4 Softener (36.0 days)

Highest days of cover

18.5 %

Pipeline share of value

33.4 %

Cycle share of value

10.5 %

Process share of value

14.5 %

Anticipation share of value

Total stock broken into its functional categories

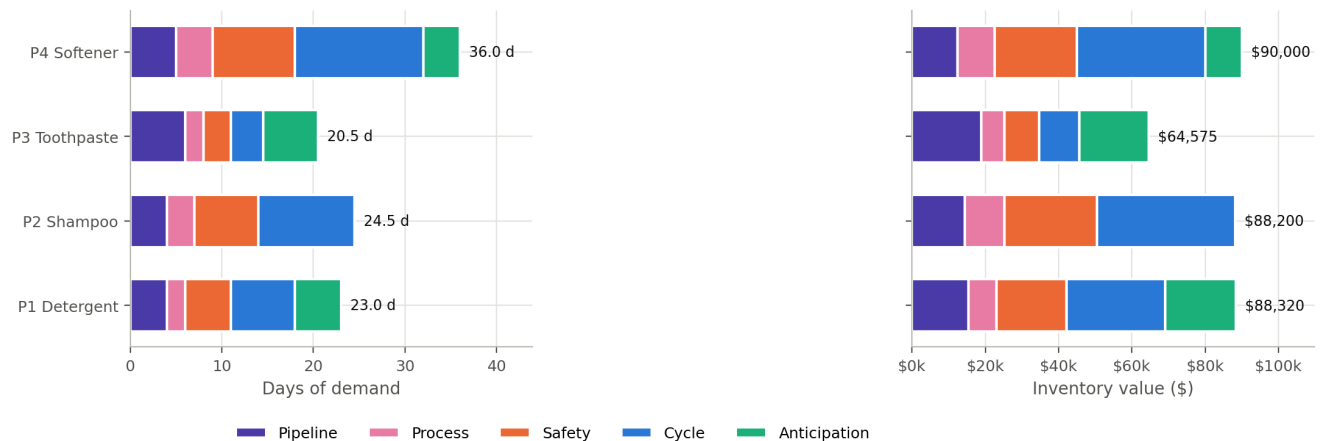


Figure 6.1 — Stacked stock components per product, in days of demand (left) and in value (right).

Stock profile over time — P1 Detergent

Pipeline, process and safety stock form a fixed base; cycle stock saw-tooths on top; anticipation stock is built ahead of a peak

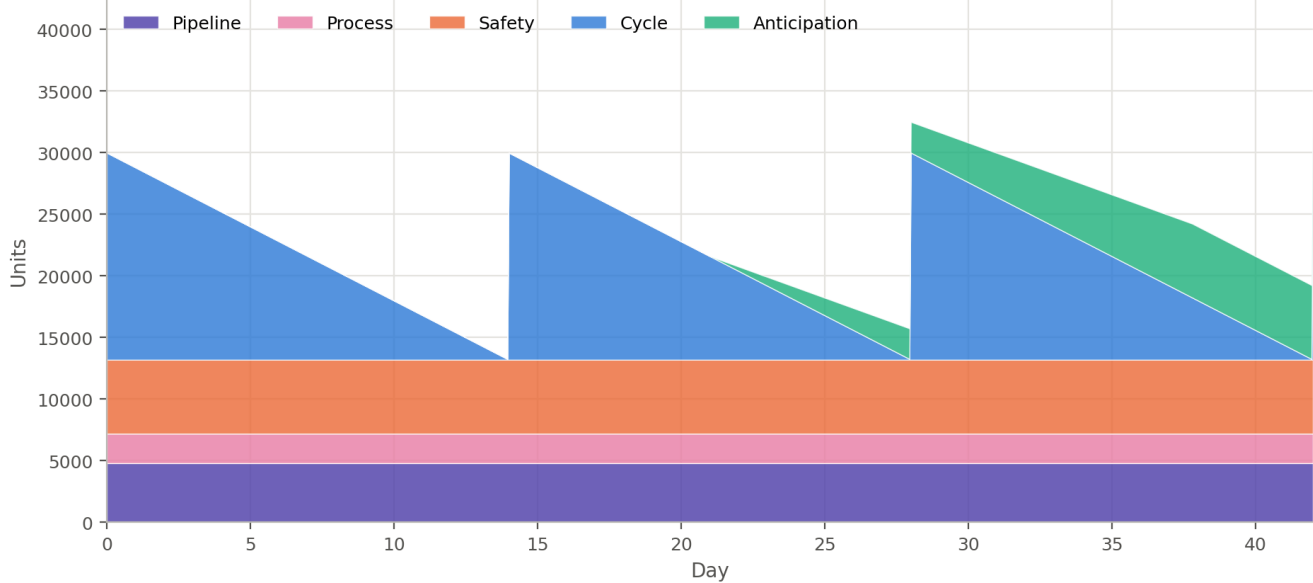


Figure 6.2 — Layered inventory profile (illustrative anticipation build in the last third of the horizon).

Cycle-stock coverage is directly proportional to the production cycle

Slope $\frac{1}{2}$: every extra day between runs adds half a day of stock; the intercept is the fixed stock

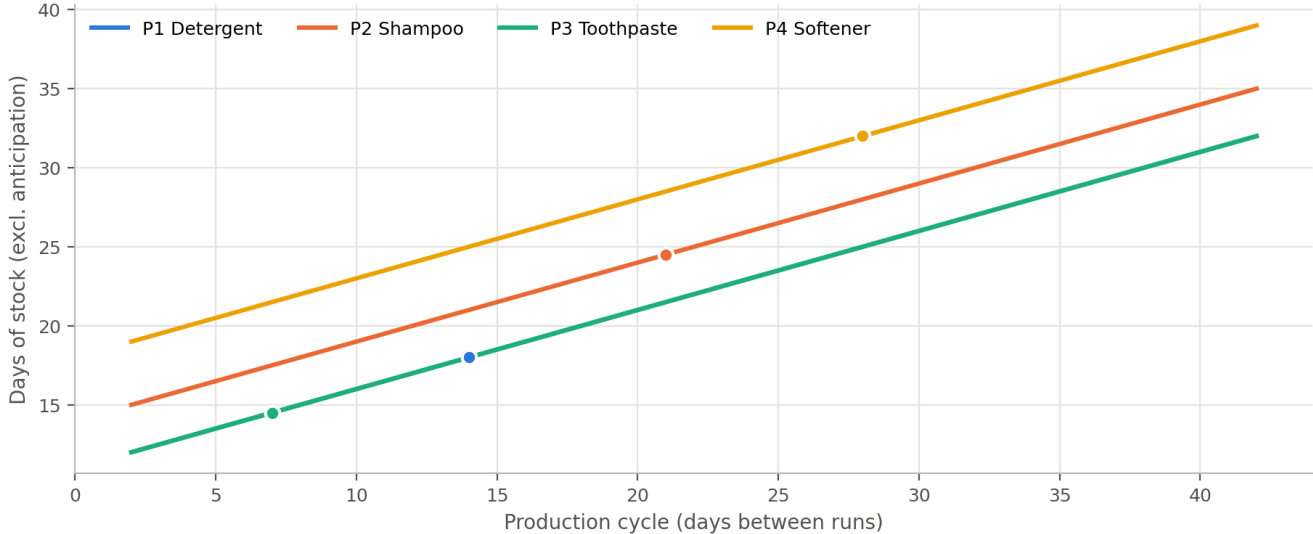


Figure 6.3 — Sensitivity of total days of stock to the production cycle length. Dots = current cycles.

Stock components in days of demand.

product	Pipeline	Process	Safety	Cycle	Anticipation	Total
P1 Detergent	4.0	2.0	5.0	7.0	5.0	23.0
P2 Shampoo	4.0	3.0	7.0	10.5	0.0	24.5
P3 Toothpaste	6.0	2.0	3.0	3.5	6.0	20.5
P4 Softener	5.0	4.0	9.0	14.0	4.0	36.0

Stock components in units.

product	Pipeline	Process	Safety	Cycle	Anticipation	Total
P1 Detergent	4,800	2,400	6,000	8,400	6,000	27,600
P2 Shampoo	3,200	2,400	5,600	8,400	0	19,600
P3 Toothpaste	9,000	3,000	4,500	5,250	9,000	30,750
P4 Softener	2,500	2,000	4,500	7,000	2,000	18,000

Stock components in value (\$).

product	Pipeline	Process	Safety	Cycle	Anticipation	Total
P1 Detergent	15,360	7,680	19,200	26,880	19,200	88,320
P2 Shampoo	14,400	10,800	25,200	37,800	0	88,200
P3 Toothpaste	18,900	6,300	9,450	11,025	18,900	64,575
P4 Softener	12,500	10,000	22,500	35,000	10,000	90,000

7. Anticipation stock: pre-building for seasonal peaks, promotions and shutdowns

Annual demand **16,000** units; regular capacity **1,250/month** (+200 overtime); August shutdown runs at **50 %** capacity.

$$E(t-1) = \max(0, D(t) - K(t) + E(t)), \quad E(12) = 0$$

INFEASIBLE — short
1,625 units

Regular capacity only

721 units

Average anticipation stock

Feasible

Regular + overtime

2,050

Overtime units needed

1,775 units (May)

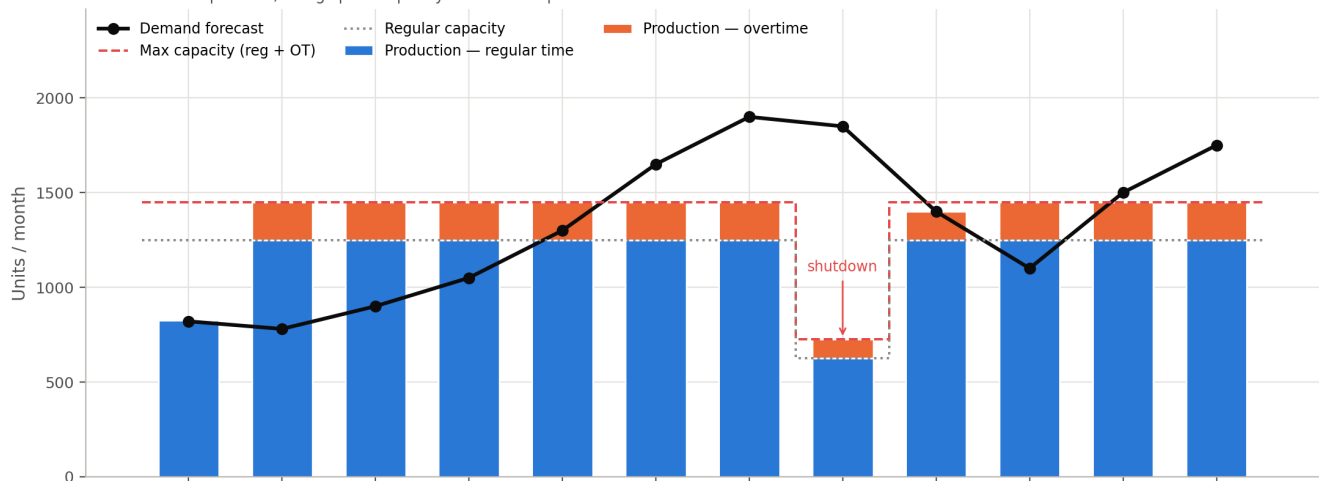
Peak anticipation stock

5

Months demand > capacity

Production plan that minimises anticipation stock

Build as late as possible, using spare capacity before each peak



Anticipation stock carried

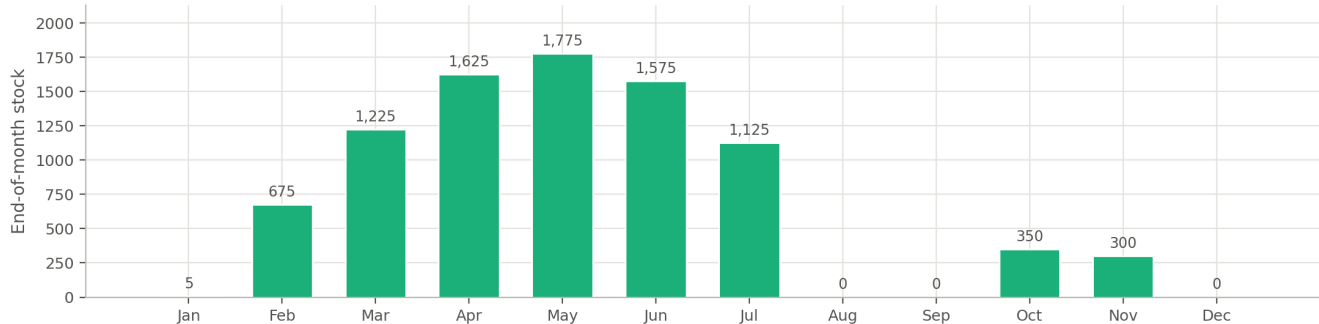


Figure 7.1 — Monthly demand vs capacity, the resulting production plan, and anticipation stock.

Cumulative view: the vertical gap is the anticipation stock

Where cumulative regular capacity falls below cumulative demand, overtime or more stock is required

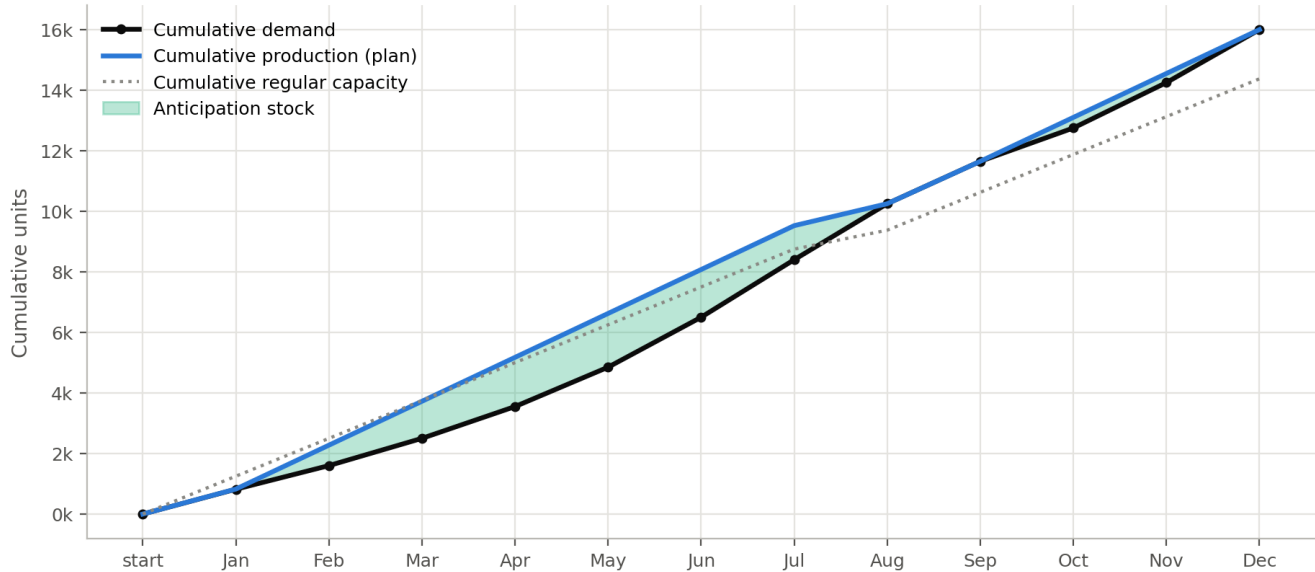


Figure 7.2 — Cumulative demand, production and regular capacity.

Monthly anticipation-stock plan (effective capacities include the shutdown).

month	demand forecast	reg cap eff	total cap eff	regular used	overtime used	production plan	anticipation stock end
Jan	820	1,250	1,450	825	0	825	5
Feb	780	1,250	1,450	1,250	200	1,450	675
Mar	900	1,250	1,450	1,250	200	1,450	1,225
Apr	1,050	1,250	1,450	1,250	200	1,450	1,625
May	1,300	1,250	1,450	1,250	200	1,450	1,775
Jun	1,650	1,250	1,450	1,250	200	1,450	1,575
Jul	1,900	1,250	1,450	1,250	200	1,450	1,125
Aug	1,850	625	725	625	100	725	0
Sep	1,400	1,250	1,450	1,250	150	1,400	0
Oct	1,100	1,250	1,450	1,250	200	1,450	350
Nov	1,500	1,250	1,450	1,250	200	1,450	300
Dec	1,750	1,250	1,450	1,250	200	1,450	0

8. Decoupling stock between two production stages

Two stages, each with mean capacity **100/day** and CV **25%**. Throughput is limited by the slower stage every day unless a buffer decouples them.

85.9 /day

Throughput, no buffer

99.7 /day

Throughput, buffer 400

16.2 %

Gain

Decoupling stock lets stages run independently

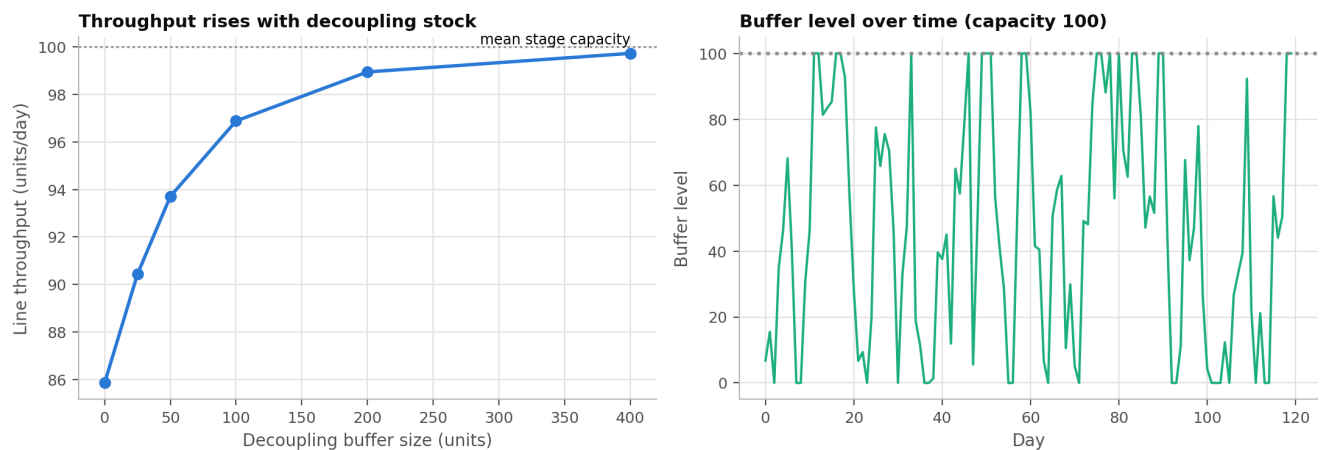


Figure 8.1 — Two-stage line simulation.

Throughput vs buffer size.

buffer capacity	throughput per day	utilisation stage2 pct	avg buffer stock
0	85.9	86.1	0.0
25	90.4	90.7	13.0
50	93.7	94.0	26.5
100	96.9	97.1	54.3
200	98.9	99.2	125.1
400	99.7	100.0	266.9

9. Why it is difficult: SKU diversity, BOMs and dependent demand

Every end item explodes into many components whose demand is **dependent** — it is *calculated* from the end-item plan, not forecast. Independent demand (end items, spare parts) must be forecast.

18 Distinct purchased components in WM-100 (model BOM)	61 Total component units per machine	2 BOM levels
5,990 12-week machine demand	365,390 12-week component units required	

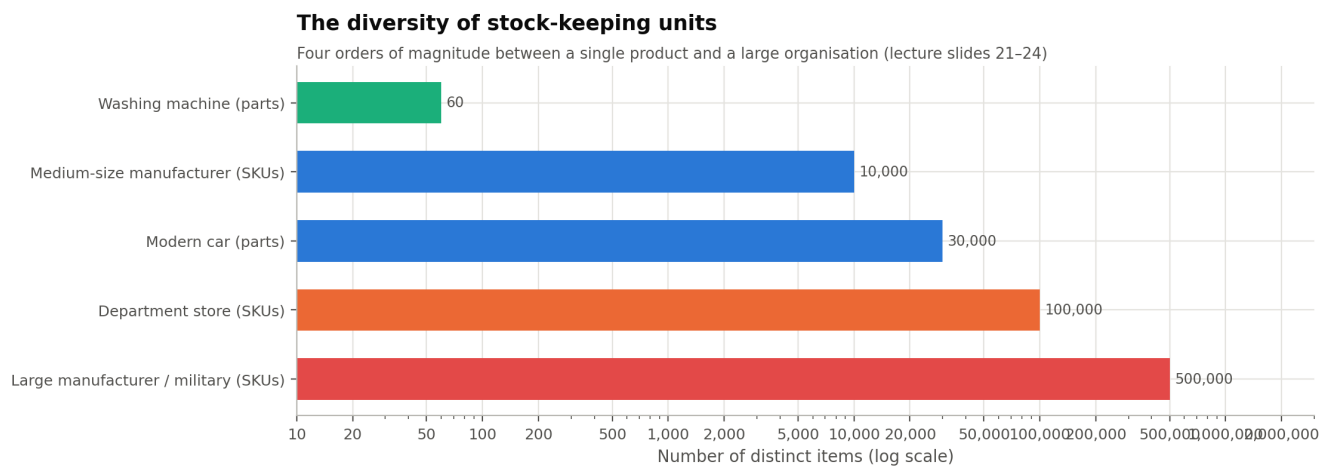


Figure 9.1 — Item counts quoted in the lecture.

Product structure (bill of materials) of a washing machine

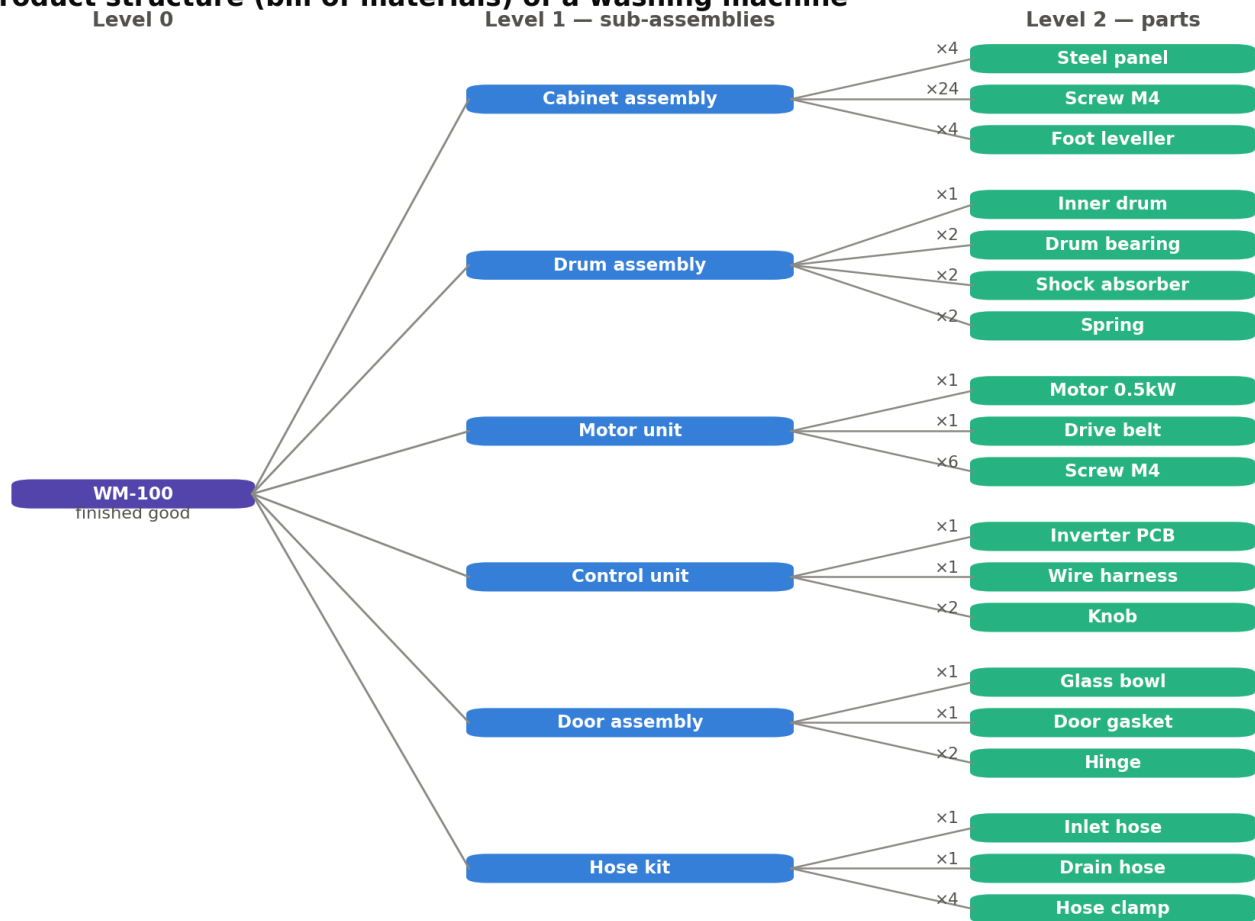


Figure 9.2 — Simplified 3-level BOM. Numbers on the links = quantity per parent.

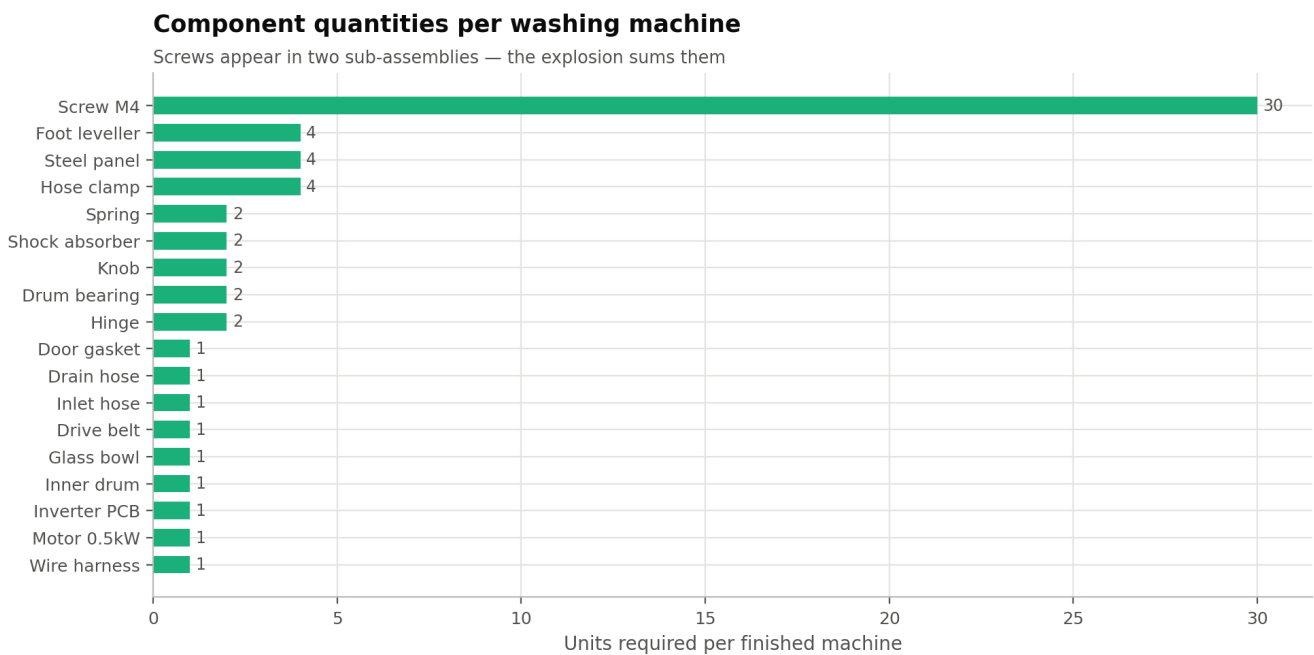


Figure 9.3 — Result of the multi-level BOM explosion.

Dependent demand: weekly gross requirements for components

Calculated from the master production schedule × BOM, not forecast

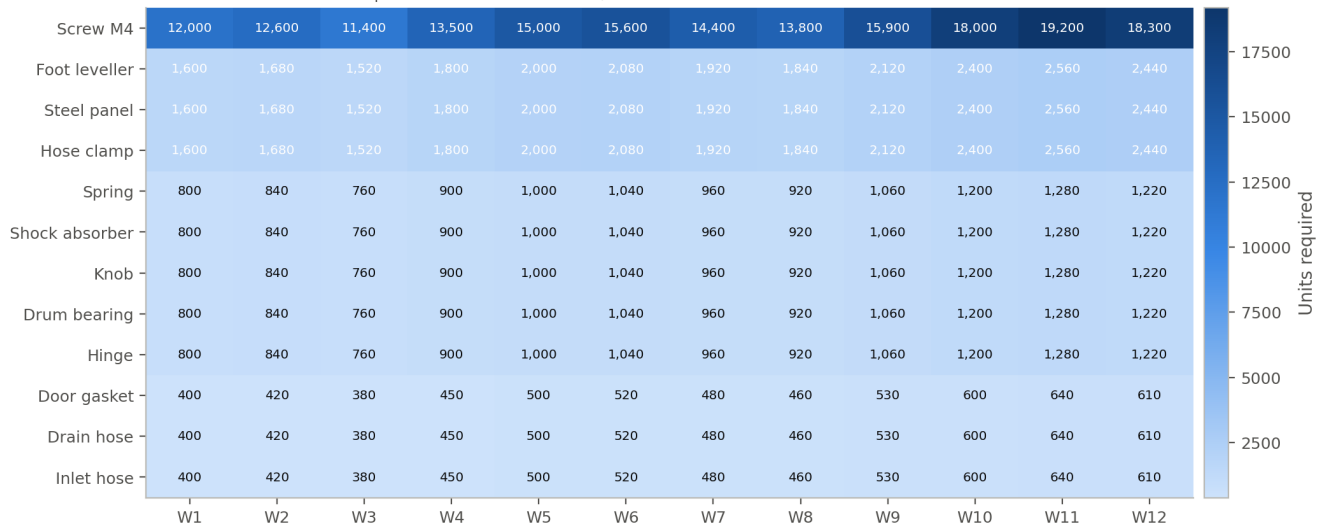


Figure 9.4 — Gross component requirements for the 12 most-used components (lead-time offsetting ignored).

Indented BOM explosion.

level	parent	component	qty per parent	qty per end item
1	WM-100	Cabinet assembly	1	1
2	Cabinet assembly	Steel panel	4	4
2	Cabinet assembly	Screw M4	24	24
2	Cabinet assembly	Foot leveller	4	4
1	WM-100	Drum assembly	1	1
2	Drum assembly	Inner drum	1	1
2	Drum assembly	Drum bearing	2	2
2	Drum assembly	Shock absorber	2	2
2	Drum assembly	Spring	2	2
1	WM-100	Motor unit	1	1
2	Motor unit	Motor 0.5kW	1	1
2	Motor unit	Drive belt	1	1
2	Motor unit	Screw M4	6	6
1	WM-100	Control unit	1	1
2	Control unit	Inverter PCB	1	1
2	Control unit	Wire harness	1	1
2	Control unit	Knob	2	2
1	WM-100	Door assembly	1	1
2	Door assembly	Glass bowl	1	1
2	Door assembly	Door gasket	1	1
2	Door assembly	Hinge	2	2
1	WM-100	Hose kit	1	1
2	Hose kit	Inlet hose	1	1
2	Hose kit	Drain hose	1	1
2	Hose kit	Hose clamp	4	4

Gross requirements by component and week.

component	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Screw M4	12,000	12,600	11,400	13,500	15,000	15,600	14,400	13,800	15,900	18,000	19,200	18,300
Foot leveller	1,600	1,680	1,520	1,800	2,000	2,080	1,920	1,840	2,120	2,400	2,560	2,440
Steel panel	1,600	1,680	1,520	1,800	2,000	2,080	1,920	1,840	2,120	2,400	2,560	2,440
Hose clamp	1,600	1,680	1,520	1,800	2,000	2,080	1,920	1,840	2,120	2,400	2,560	2,440
Spring	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Shock absorber	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Knob	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Drum bearing	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Hinge	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Door gasket	400	420	380	450	500	520	480	460	530	600	640	610
Drain hose	400	420	380	450	500	520	480	460	530	600	640	610
Inlet hose	400	420	380	450	500	520	480	460	530	600	640	610
Drive belt	400	420	380	450	500	520	480	460	530	600	640	610
Glass bowl	400	420	380	450	500	520	480	460	530	600	640	610
Inner drum	400	420	380	450	500	520	480	460	530	600	640	610
Inverter PCB	400	420	380	450	500	520	480	460	530	600	640	610
Motor 0.5kW	400	420	380	450	500	520	480	460	530	600	640	610
Wire harness	400	420	380	450	500	520	480	460	530	600	640	610

10. Product variety multiplies the stock needed for the same service

$$SS_{total}(n) = \sqrt{n} \times SS_{total}(1)$$

1,234 units

SS with 1 variant

\$63,908/yr

Extra holding cost at max variety

9,870 units (×8.0)

SS with 64 variants

95.0% / 95.0%

Target / simulated CSL

The cost of variety: the square-root law

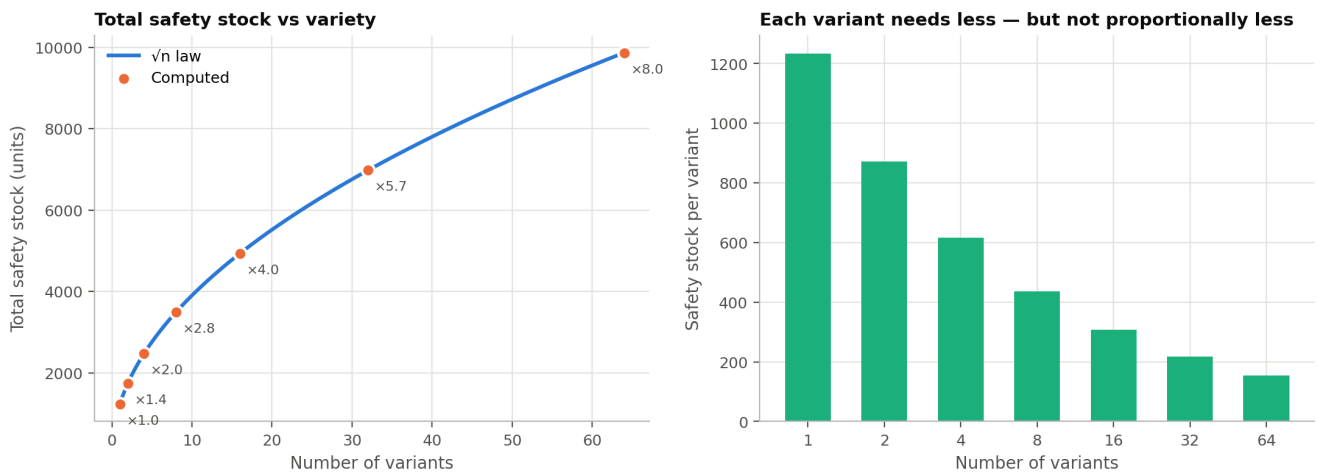


Figure 10.1 — Total demand 1,000/day, CV 25%, lead time 9 days, $z = 1.645$.

Safety stock as variety grows.

n variants	mean daily demand per variant	sigma per variant	SS per variant	SS total	SS total index	simulated CSL	annual holding cost SS
1	1,000.0	250.0	1,233.8	1,234	1.00	95.5%	9,130
2	500.0	176.8	872.4	1,745	1.41	94.6%	12,911
4	250.0	125.0	616.9	2,468	2.00	95.1%	18,260
8	125.0	88.4	436.2	3,490	2.83	95.0%	25,823
16	62.5	62.5	308.4	4,935	4.00	95.0%	36,519
32	31.2	44.2	218.1	6,979	5.66	95.0%	51,646
64	15.6	31.2	154.2	9,870	8.00	94.9%	73,038

- Promotions create short demand spikes → anticipation stock or capacity costs (Parts 2.4, 6.6).
- Variety should be priced with its inventory consequence; *postponement* (customising late) pools demand.

11. ABC analysis — lecture example (10 part numbers)

Step 1 — annual dollar usage

Annual usage analysis (slide 32).

part number	annual unit usage	unit cost	annual dollar usage
1	1,100	\$2	\$2,200
2	600	\$40	\$24,000
3	100	\$4	\$400
4	1,300	\$1	\$1,300
5	100	\$60	\$6,000
6	10	\$25	\$250
7	100	\$2	\$200
8	1,500	\$2	\$3,000
9	200	\$2	\$400
10	500	\$1	\$500

\$38,250

Total annual \$ usage

10

Number of parts

Step 2 — rank, cumulate and classify

Ranking by annual \$ usage (slide 34) with classes from both cut-off rules.

part number	annual dollar usage	cum value	cum pct value	cum pct items	class	class by value 80 95
2	\$24,000	\$24,000	62.7%	10%	A	A
5	\$6,000	\$30,000	78.4%	20%	A	A
8	\$3,000	\$33,000	86.3%	30%	B	B
1	\$2,200	\$35,200	92.0%	40%	B	B
4	\$1,300	\$36,500	95.4%	50%	B	C
10	\$500	\$37,000	96.7%	60%	C	C
9	\$400	\$37,400	97.8%	70%	C	C
3	\$400	\$37,800	98.8%	80%	C	C
6	\$250	\$38,050	99.5%	90%	C	C
7	\$200	\$38,250	100.0%	100%	C	C

With the **items rule (20 % / 30 % / 50 %)** parts 2 and 5 are A, parts 8, 1, 4 are B and the rest C — A holds **78 %** of the value with 20 % of the items, exactly the Pareto pattern on slide 30. With **strict value cut-offs (80 % / 95 %)** part 4 (cumulative 95.4 %) falls just into C: items at a boundary are a **judgement call** — managers often promote them for other reasons (criticality, scarcity).

Pareto chart — lecture example

Part 2 alone is 63 % of annual \$ usage

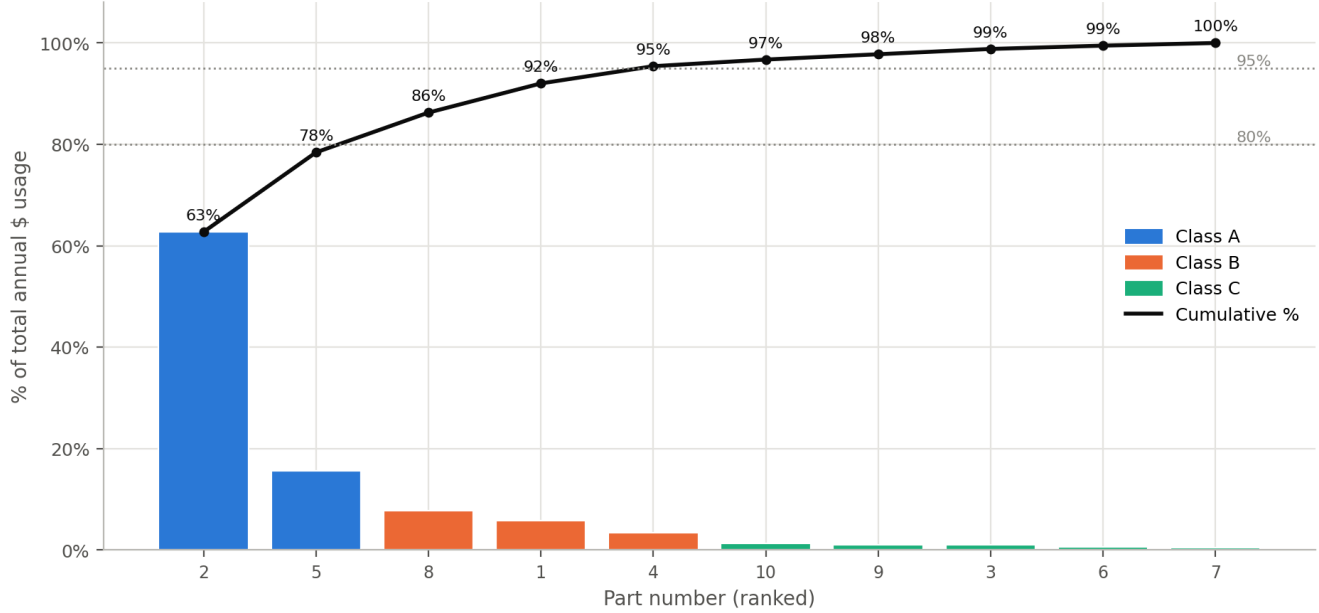


Figure 11.1 — Bars = % of annual \$ usage per part (coloured by class); line = cumulative %.

ABC curve

Cumulative % of items vs cumulative % of value

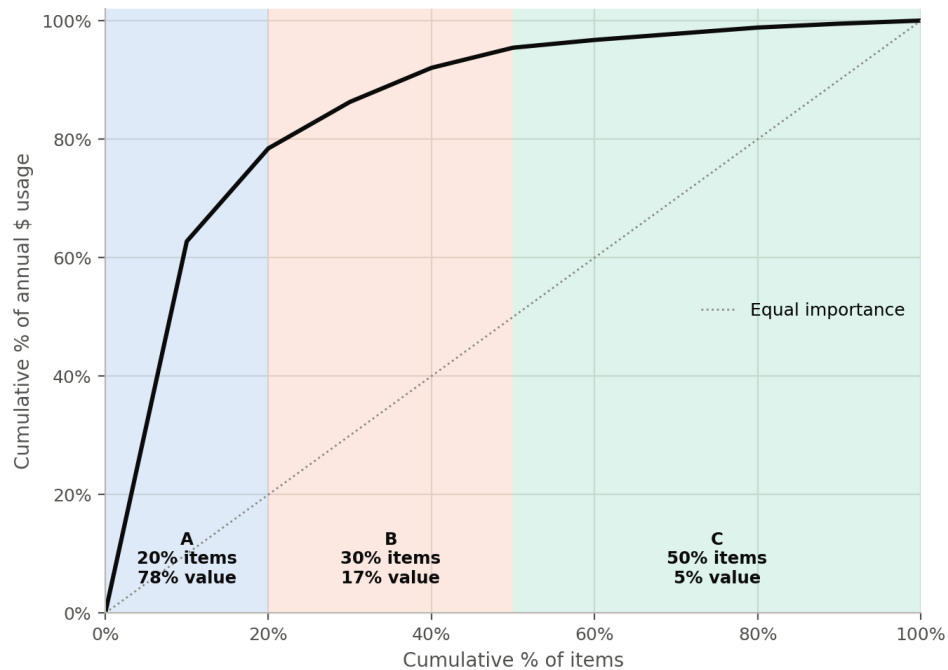


Figure 11.2 — ABC (Lorenz) curve with class bands.

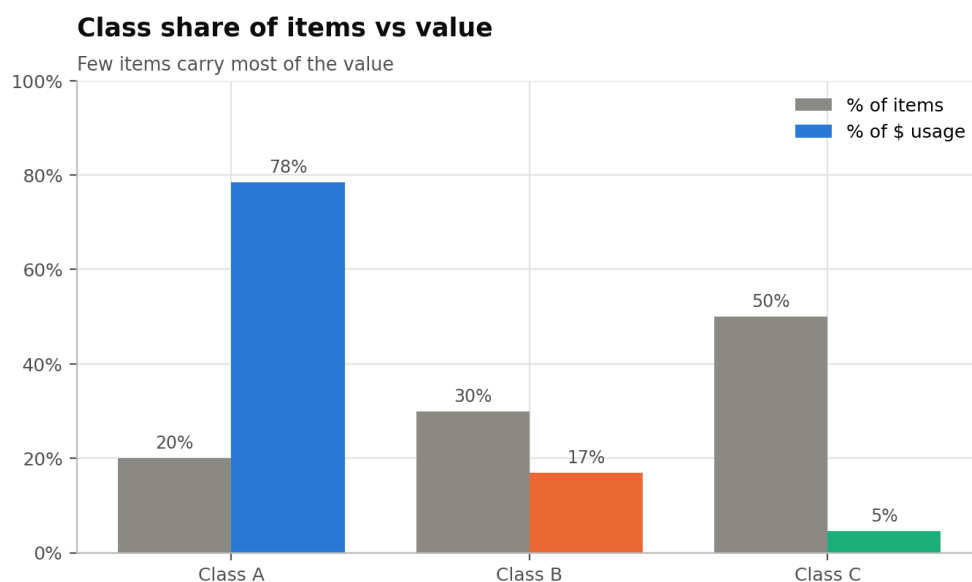


Figure 11.3 — Share of items and value by class.

Class summary (items rule).

class	items	value	pct items	pct value
A	2	\$30,000	20%	78.4%
B	3	\$6,500	30%	17.0%
C	5	\$1,750	50%	4.6%

12. ABC classification of 200 SKUs

Classification rule: cumulative % of value — A ≤ 80%, B ≤ 95%, C = rest.

\$1,832,644

Total annual \$ usage

**91 items (46%) →
5.1% of value**

Class C

**45 items (22%) →
79.8% of value**

Class A

**SKU-0172 = 19.4% of
value**

Top SKU

**64 items (32%) →
15.1% of value**

Class B

0.75

Concentration (Gini)

Pareto chart — 200 SKUs

45 A-items (22%) account for 80% of annual \$ usage

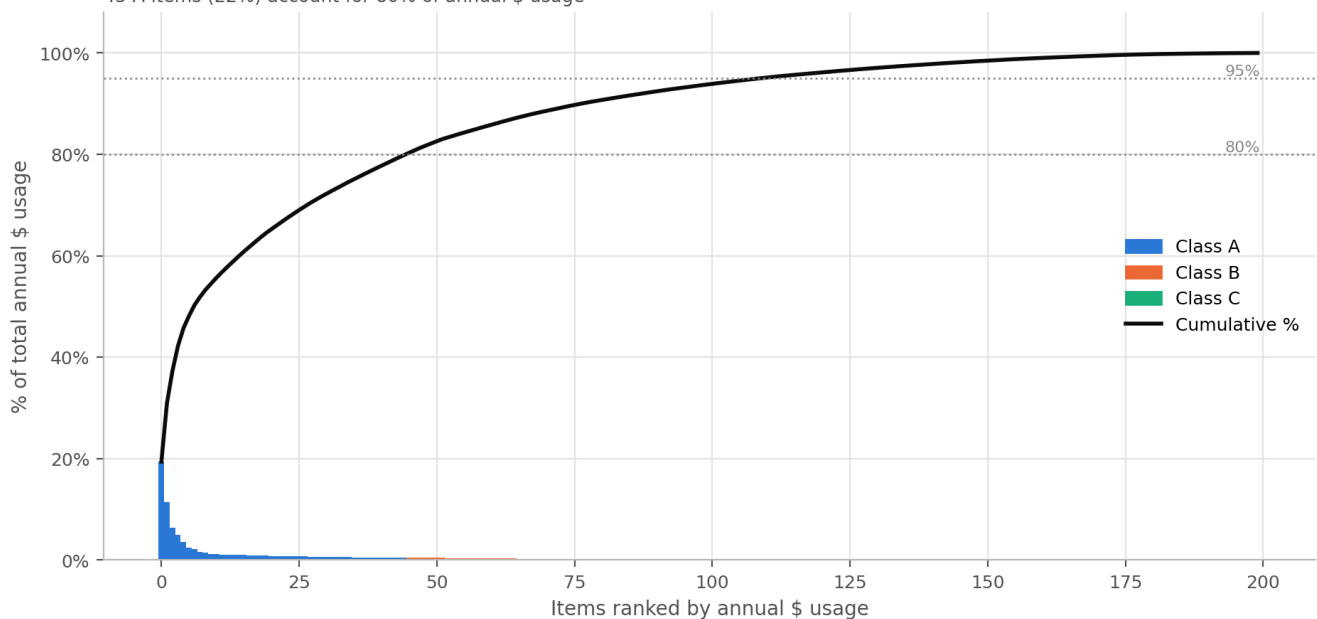


Figure 12.1 — Pareto chart of all 200 SKUs coloured by class.

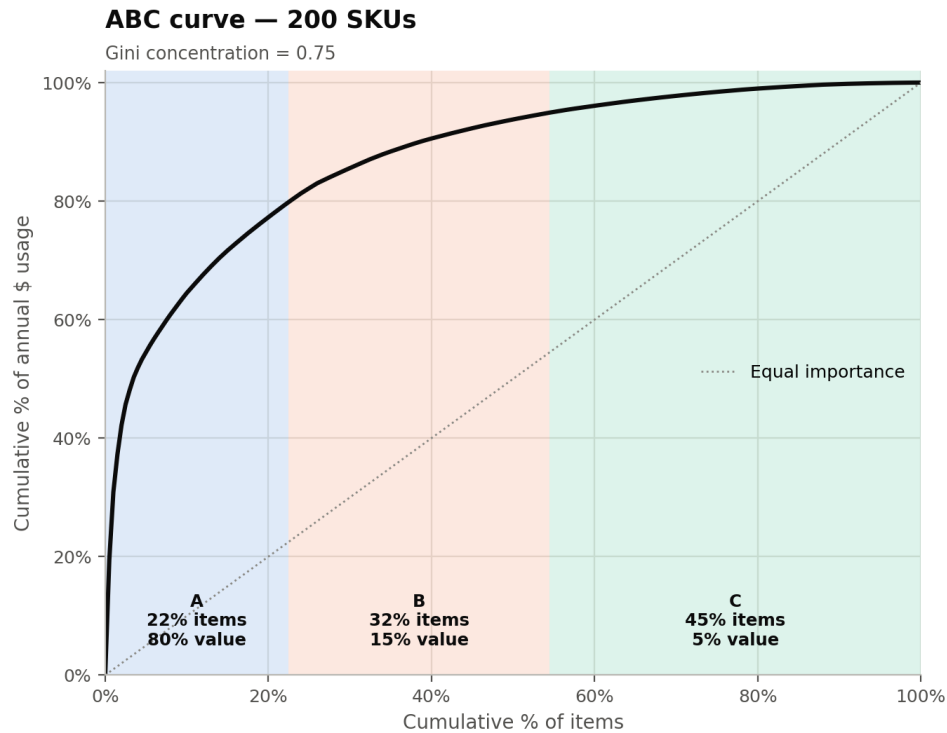
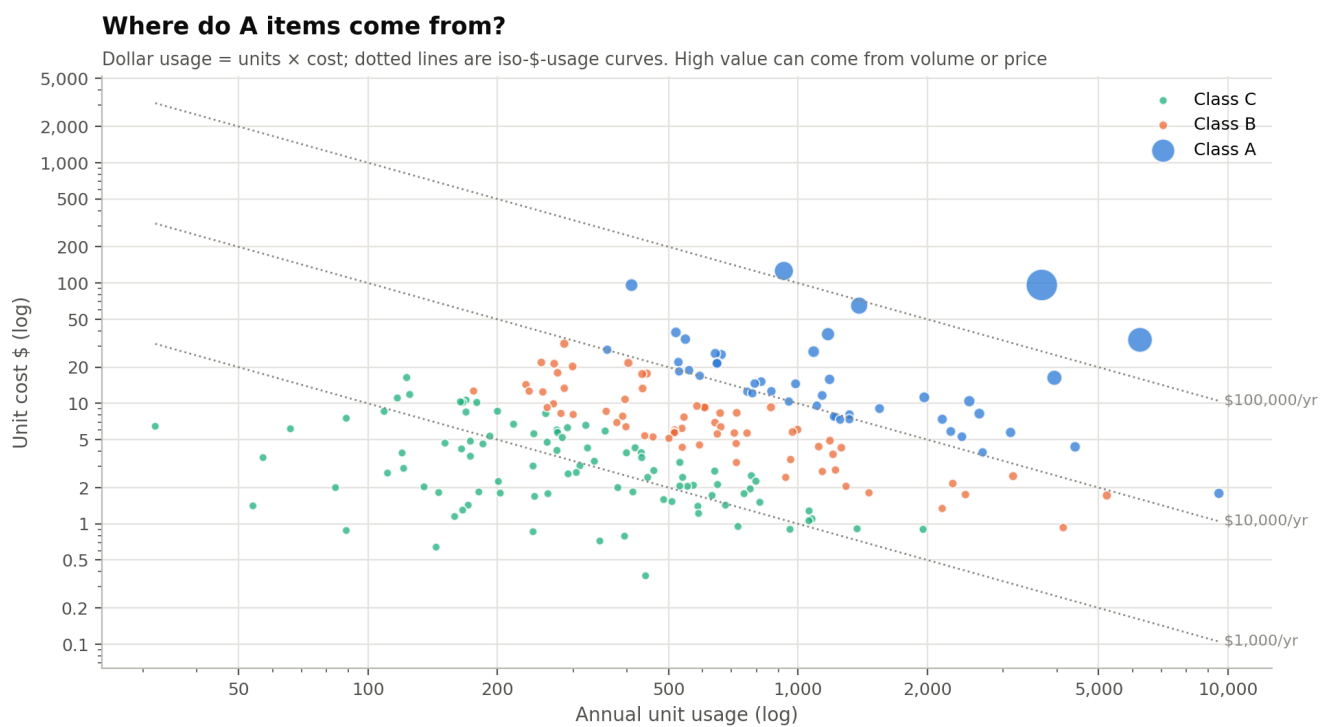


Figure 12.2 — ABC (Lorenz) curve.

Figure 12.3 — Each bubble is a SKU; bubble area $\propto$ annual \$ usage.

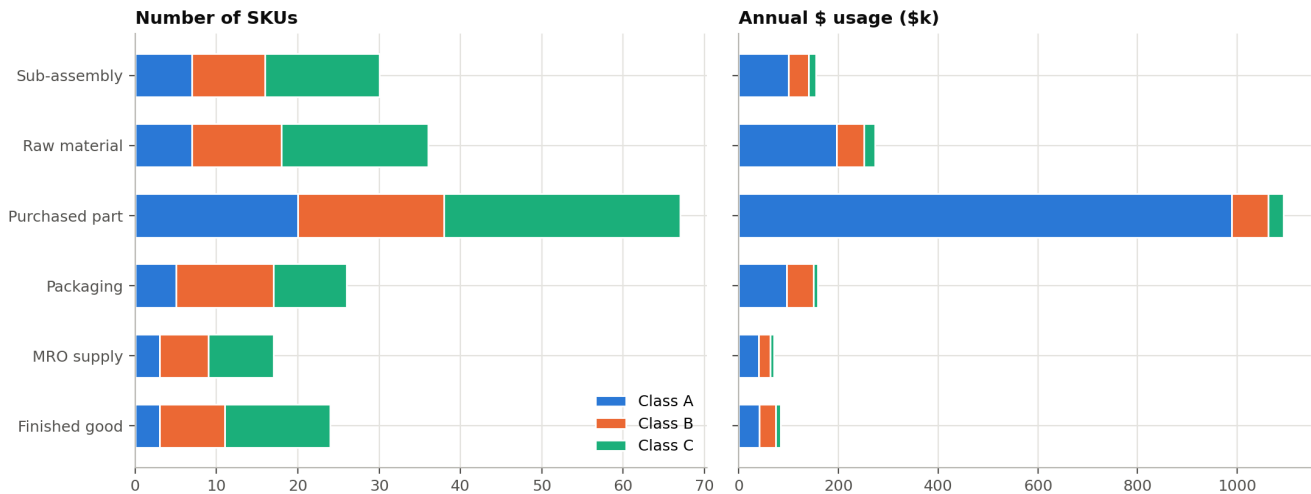
ABC mix by item category

Figure 12.4 — Count and value of SKUs by category and class.

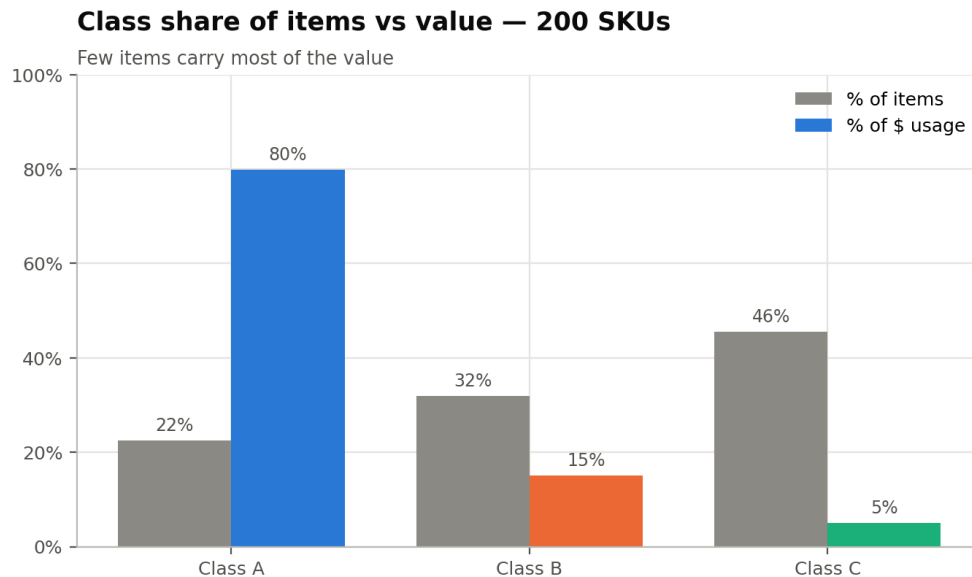


Figure 12.5 — Class summary.

Class summary.

class	items	value	pct items	pct value
A	45	\$1,463,320	22.5%	79.8%
B	64	\$276,463	32.0%	15.1%
C	91	\$92,861	45.5%	5.1%

Ranked SKU list (first 40 rows; full list in reports/tables).

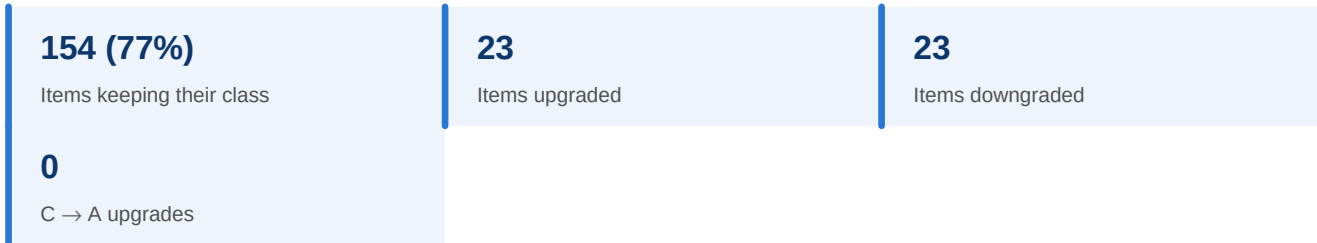
rank	sku	category	annual unit usage	unit cost	annual dollar usage	pct of value	cum pct value	cum pct items	class
1	SKU-0172	Purchased part	3,689	\$96.42	\$355,693	19.41%	19.4%	0.5%	A
2	SKU-0052	Purchased part	6,240	\$33.75	\$210,600	11.49%	30.9%	1.0%	A
3	SKU-0123	Purchased part	927	\$126.14	\$116,932	6.38%	37.3%	1.5%	A
4	SKU-0176	Raw material	1,387	\$64.94	\$90,072	4.91%	42.2%	2.0%	A
5	SKU-0046	Purchased part	3,946	\$16.34	\$64,478	3.52%	45.7%	2.5%	A
6	SKU-0024	Raw material	1,174	\$37.61	\$44,154	2.41%	48.1%	3.0%	A

rank	sku	category	annual unit usage	unit cost	annual dollar usage	pct of value	cum pct value	cum pct items	class
7	SKU-0041	Packaging	410	\$96.15	\$39,422	2.15%	50.3%	3.5%	A
8	SKU-0019	Purchased part	1,087	\$26.94	\$29,284	1.60%	51.9%	4.0%	A
9	SKU-0043	Purchased part	2,502	\$10.44	\$26,121	1.43%	53.3%	4.5%	A
10	SKU-0192	Sub-assembly	1,966	\$11.24	\$22,098	1.21%	54.5%	5.0%	A
11	SKU-0117	Raw material	2,640	\$8.23	\$21,727	1.19%	55.7%	5.5%	A
12	SKU-0112	Purchased part	520	\$38.94	\$20,249	1.10%	56.8%	6.0%	A
13	SKU-0069	Packaging	4,409	\$4.36	\$19,223	1.05%	57.8%	6.5%	A
14	SKU-0130	Sub-assembly	1,184	\$15.85	\$18,766	1.02%	58.9%	7.0%	A
15	SKU-0072	Purchased part	547	\$34.22	\$18,718	1.02%	59.9%	7.5%	A
16	SKU-0185	Purchased part	3,121	\$5.74	\$17,915	0.98%	60.9%	8.0%	A
17	SKU-0109	Sub-assembly	9,523	\$1.79	\$17,046	0.93%	61.8%	8.5%	A
18	SKU-0134	Finished good	663	\$25.46	\$16,880	0.92%	62.7%	9.0%	A
19	SKU-0016	MRO supply	642	\$25.97	\$16,673	0.91%	63.6%	9.5%	A
20	SKU-0045	Purchased part	2,165	\$7.39	\$15,999	0.87%	64.5%	10.0%	A
21	SKU-0082	Packaging	987	\$14.56	\$14,371	0.78%	65.3%	10.5%	A
22	SKU-0111	Packaging	649	\$21.75	\$14,116	0.77%	66.1%	11.0%	A
23	SKU-0091	Finished good	1,547	\$9.07	\$14,031	0.77%	66.8%	11.5%	A
24	SKU-0005	Purchased part	648	\$21.52	\$13,945	0.76%	67.6%	12.0%	A
25	SKU-0158	Purchased part	1,138	\$11.65	\$13,258	0.72%	68.3%	12.5%	A
26	SKU-0083	Purchased part	2,266	\$5.85	\$13,256	0.72%	69.0%	13.0%	A
27	SKU-0141	Sub-assembly	2,405	\$5.29	\$12,722	0.69%	69.7%	13.5%	A
28	SKU-0179	MRO supply	821	\$15.18	\$12,463	0.68%	70.4%	14.0%	A
29	SKU-0099	Purchased part	527	\$22.04	\$11,615	0.63%	71.0%	14.5%	A
30	SKU-0085	Raw material	793	\$14.63	\$11,602	0.63%	71.7%	15.0%	A
31	SKU-0200	Purchased part	867	\$12.63	\$10,950	0.60%	72.3%	15.5%	A
32	SKU-0053	Purchased part	1,317	\$8.10	\$10,668	0.58%	72.8%	16.0%	A
33	SKU-0110	Purchased part	1,106	\$9.58	\$10,595	0.58%	73.4%	16.5%	A
34	SKU-0002	Sub-assembly	558	\$18.90	\$10,546	0.58%	74.0%	17.0%	A
35	SKU-0164	MRO supply	2,687	\$3.92	\$10,533	0.57%	74.6%	17.5%	A
36	SKU-0156	Raw material	360	\$27.93	\$10,055	0.55%	75.1%	18.0%	A
37	SKU-0066	Finished good	591	\$16.97	\$10,029	0.55%	75.7%	18.5%	A
38	SKU-0054	Purchased part	952	\$10.37	\$9,872	0.54%	76.2%	19.0%	A
39	SKU-0026	Purchased part	529	\$18.45	\$9,760	0.53%	76.7%	19.5%	A
40	SKU-0097	Sub-assembly	1,316	\$7.40	\$9,738	0.53%	77.3%	20.0%	A

... 160 more rows not shown — full table exported to reports/tables/.

13. Multi-criteria ABC: dollar usage, scarcity and quality problems

Weights: dollar usage **0.6**, scarcity **0.25**, quality **0.15**. Both classifications use the 20/30/50 % items rule so that they are comparable.



Adding scarcity and quality to the ABC criteria

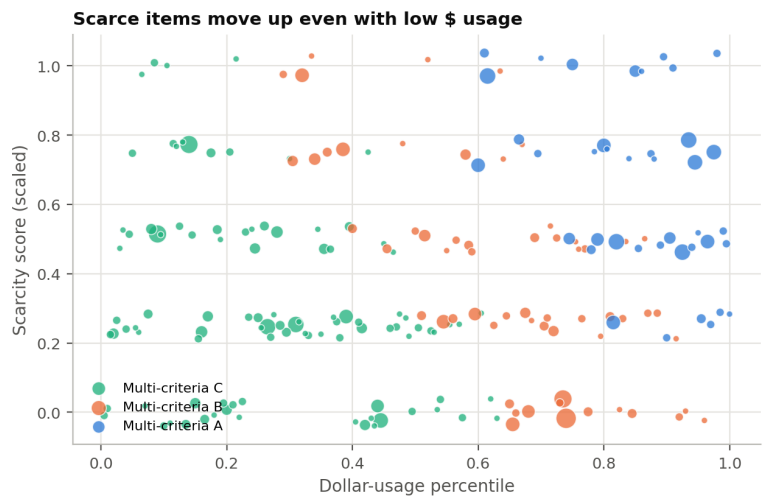
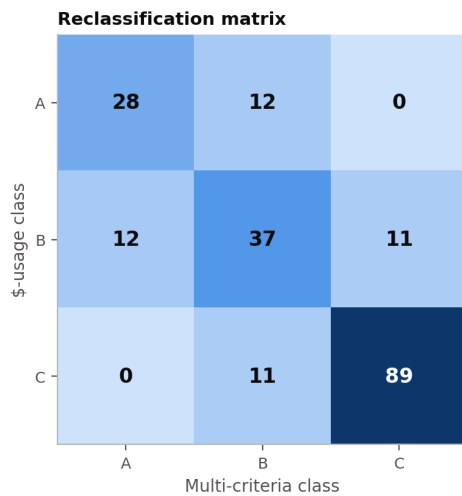


Figure 13.1 — Left: counts moving between classes. Right: bubble size $\propto$ quality incidents.

Largest upgrades (items that deserve more control than \$ usage alone suggests).

sku	category	annual dollar usage	scarcity score 1to5	quality incidents per year	score	class dollar	class multi
SKU-0167	Finished good	\$7,876	5	4	0.828	B	A
SKU-0155	Packaging	\$9,548	4	6	0.805	B	A
SKU-0034	Sub-assembly	\$3,821	5	8	0.764	B	A
SKU-0186	Sub-assembly	\$9,489	3	5	0.731	B	A
SKU-0175	Finished good	\$3,621	5	2	0.721	B	A
SKU-0070	Purchased part	\$4,892	4	3	0.706	B	A
SKU-0086	Sub-assembly	\$7,617	3	4	0.700	B	A
SKU-0017	Purchased part	\$9,184	3	2	0.698	B	A
SKU-0126	Raw material	\$5,618	5	0	0.695	B	A
SKU-0023	Packaging	\$3,339	4	6	0.685	B	A
SKU-0106	Raw material	\$9,417	4	0	0.684	B	A
SKU-0077	Raw material	\$5,541	4	1	0.677	B	A
SKU-0108	MRO supply	\$1,378	5	6	0.579	C	B
SKU-0145	Finished good	\$1,693	4	6	0.556	C	B
SKU-0149	Raw material	\$1,474	4	4	0.519	C	B

Reclassification matrix.

\$-usage class \ multi-criteria class	A	B	C
A	28	12	0
B	12	37	11
C	0	11	89

14. Control policies by ABC class

Recommended control policy by class.

class	level of control	review	records	safety stock and lot size	who manages
A	Tight	Continuous / weekly	Perpetual, audited, cycle-counted monthly	Low SS, small frequent lots, close follow-up	Senior planner
B	Normal	Bi-weekly to monthly	Perpetual, counted quarterly	Moderate	Planner
C	Least possible	Quarterly or two-bin	Simple, counted yearly	Generous SS, large infrequent lots	Clerk / automatic

Control effort follows value

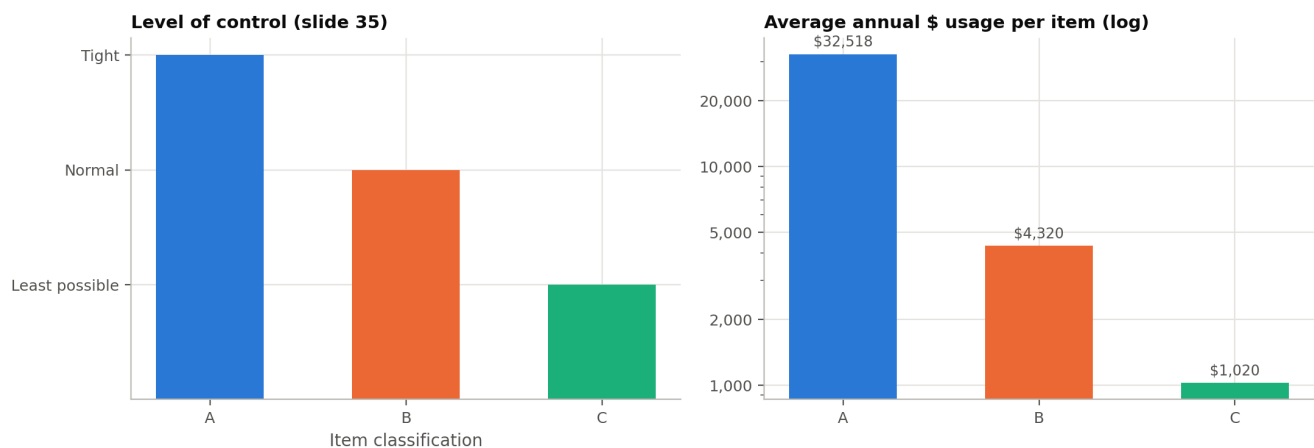


Figure 14.1 — Control level and value per item by class.

An A item is on average **32×** more valuable per year than a C item — a stock-out of a C item is an inconvenience that is cheaply prevented by holding more of it.

15. Inventory performance: \$ value, turns and days of supply

Lecture worked examples

Turns = $1,000,000 / 500,000 = 2$ Days of supply = $6,000 / 200 = 30$ days
 $\rightarrow 360/30 = 12$ turns

2

Turns (COGS example)

30 days

Days of supply (unit example)

12 per year

Equivalent turns

Division benchmark

Average inventory is the mean of the opening and four quarter-end balances (a better estimate than a single year-end snapshot, which can be window-dressed).

6.05

Company-wide turns

60 days

Company-wide days of supply

Automotive Components (36.0 turns)

Best division

Automotive Components (-4.0)

Worst vs benchmark

\$42.4 M

Total \$ inventory (Q4)

\$0.93 M/yr

Carrying-cost gap vs benchmarks

\$ value of inventory held (year-end snapshot)

Balance-sheet inventory = RM + WIP + FG

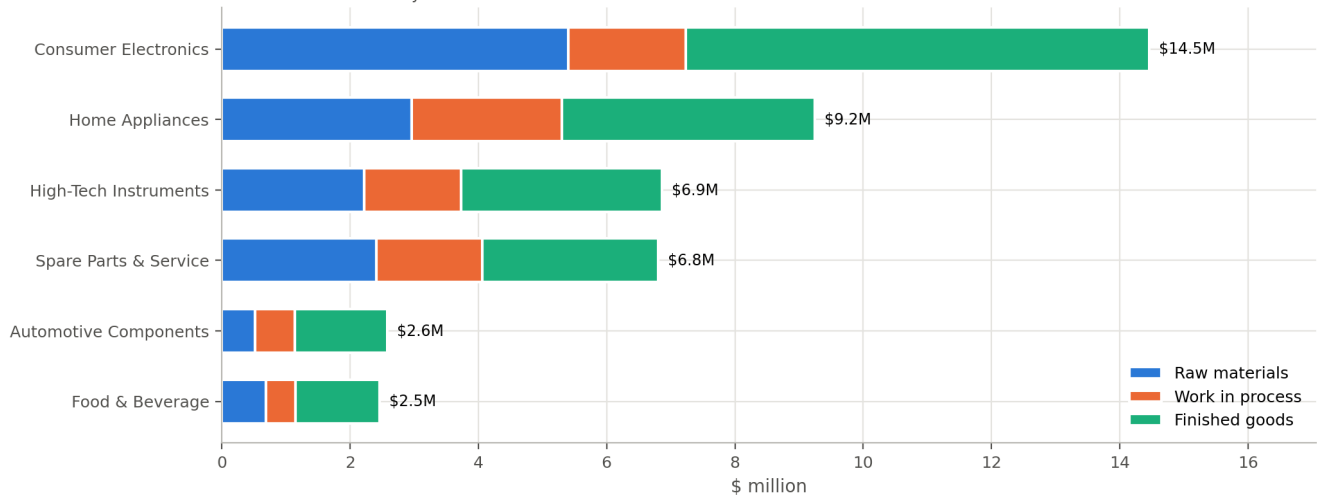


Figure 15.1 — Year-end inventory by division and stage.

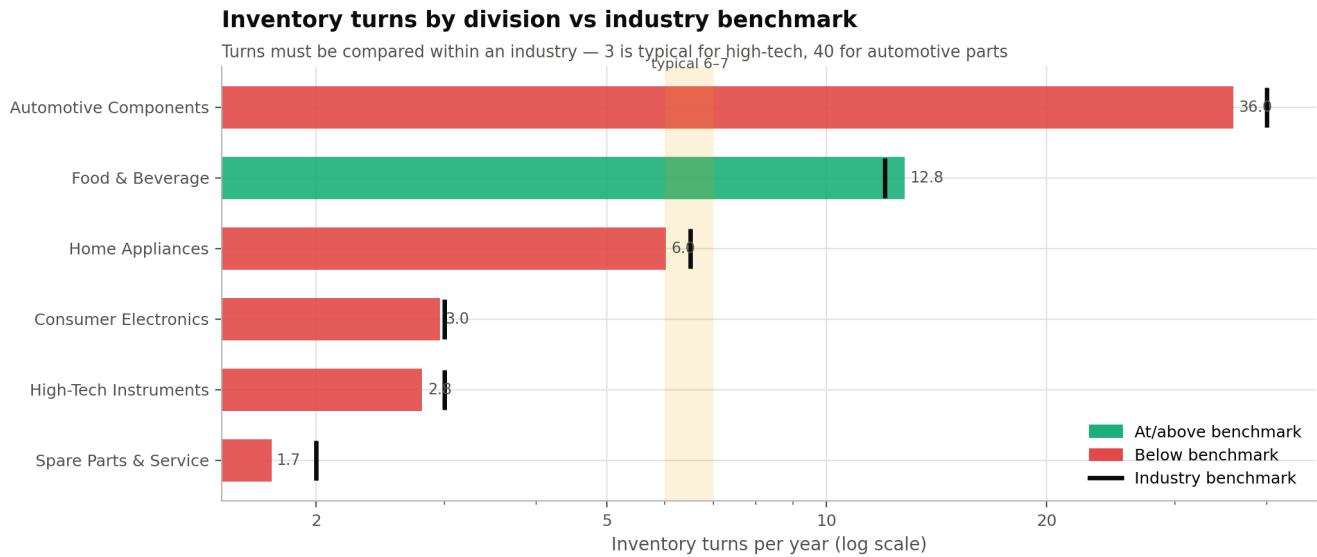


Figure 15.2 — Turns = annual COGS / average inventory.

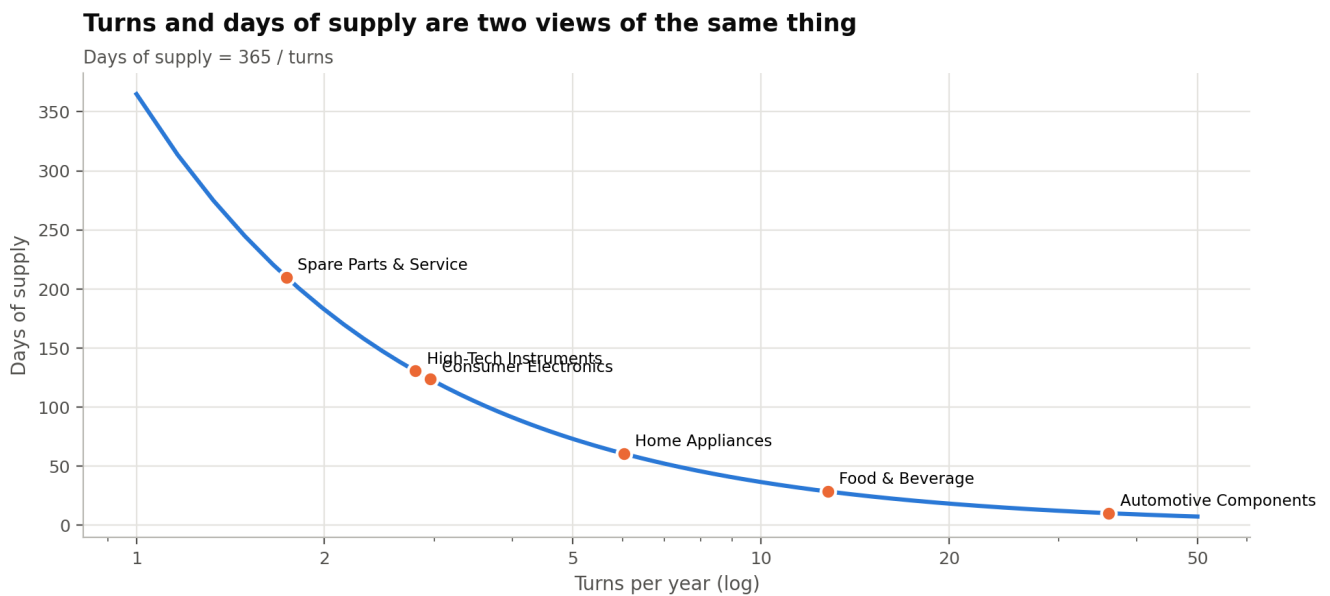


Figure 15.3 — The hyperbola links turns and days of supply.

Quarter-end inventory relative to each division's average

Why average inventory beats a single snapshot: balances swing $\pm 15\%$ during the year

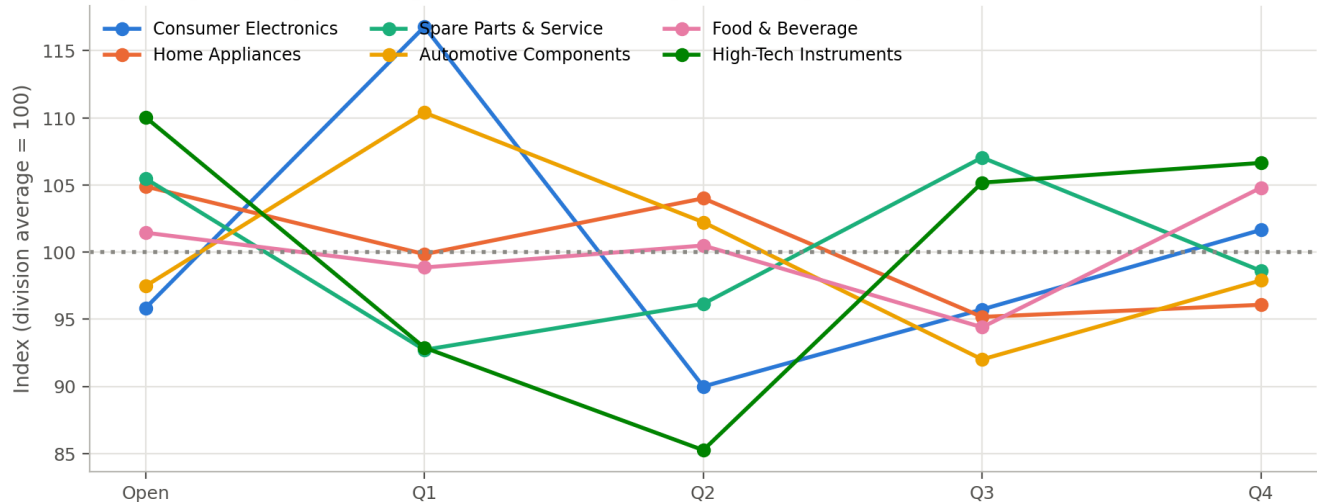


Figure 15.4 — Quarter-end balances indexed to the division's average.

Division turnover analysis.

division	annual cogs	avg inventory	year end inventory	turns	turns on year end	days of supply	industry benchmark turns	turns gap vs benchmark	excess carrying cost
Consumer Electronics	\$42,000,000	\$14,222,563	\$14,457,599	2.95	2.91	124	3.0	-0.05	\$82,348
Home Appliances	\$58,000,000	\$9,617,652	\$9,240,231	6.03	6.28	61	6.5	-0.47	\$256,993
Spare Parts & Service	\$12,000,000	\$6,904,519	\$6,808,137	1.74	1.76	210	2.0	-0.26	\$334,672
Automotive Components	\$95,000,000	\$2,636,740	\$2,581,215	36.03	36.80	10	40.0	-3.97	\$96,844
Food & Beverage	\$30,000,000	\$2,346,218	\$2,458,859	12.79	12.20	29	12.0	+0.79	-\$56,899
High-Tech Instruments	\$18,000,000	\$6,439,082	\$6,867,108	2.80	2.62	131	3.0	-0.20	\$162,460

Item-level days of supply

Days of supply by item

ITEM-00 is the lecture example: 6,000 units / 200 per day = 30 days

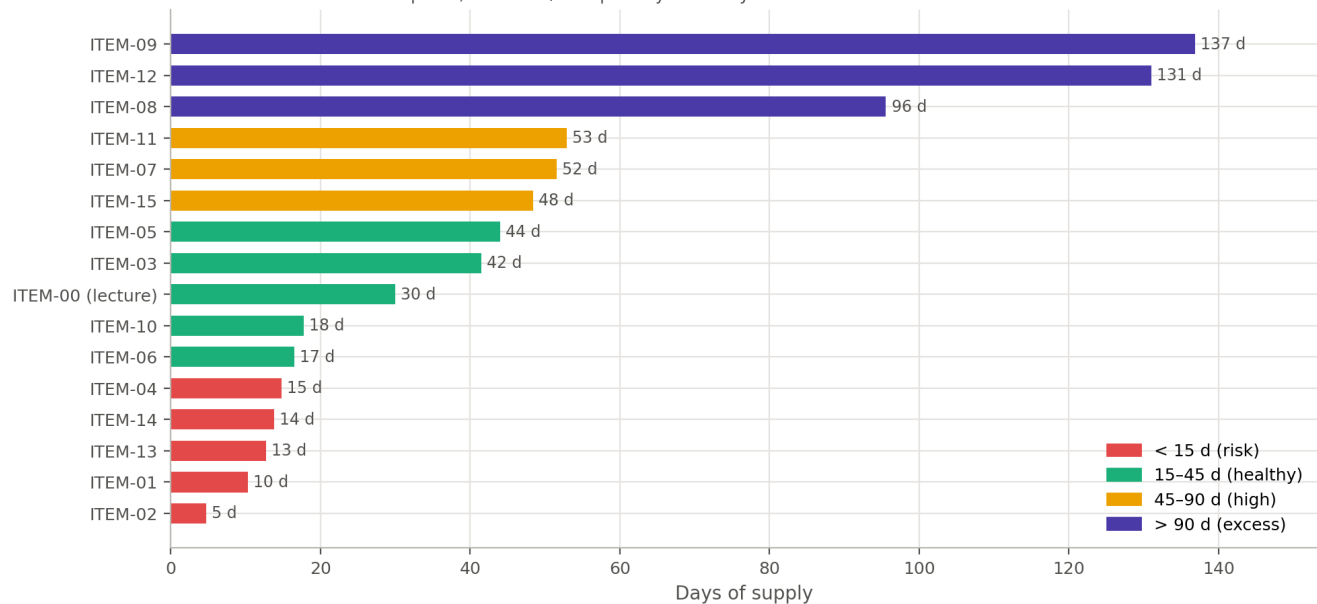


Figure 15.5 — Days of supply = inventory on hand / average daily usage.

Item days of supply and equivalent turns.

item	inventory on hand units	avg daily usage units	unit cost	days of supply	turns equiv	inventory value	band
ITEM-00 (lecture)	6,000	200	\$12.00	30.0	12.0	\$72,000	15–45 d (healthy)
ITEM-01	3,735	362	\$2.31	10.3	34.9	\$8,628	< 15 d (risk)
ITEM-02	1,343	283	\$4.28	4.7	75.9	\$5,748	< 15 d (risk)
ITEM-03	15,651	377	\$4.74	41.5	8.7	\$74,186	15–45 d (healthy)
ITEM-04	593	40	\$10.63	14.8	24.3	\$6,304	< 15 d (risk)
ITEM-05	11,396	259	\$4.93	44.0	8.2	\$56,182	15–45 d (healthy)
ITEM-06	2,662	161	\$4.54	16.5	21.8	\$12,085	15–45 d (healthy)
ITEM-07	7,834	152	\$8.66	51.5	7.0	\$67,842	45–90 d (high)
ITEM-08	13,666	143	\$5.94	95.6	3.8	\$81,176	> 90 d (excess)
ITEM-09	30,101	220	\$13.52	136.8	2.6	\$406,966	> 90 d (excess)
ITEM-10	373	21	\$18.65	17.8	20.3	\$6,956	15–45 d (healthy)
ITEM-11	13,644	258	\$7.92	52.9	6.8	\$108,060	45–90 d (high)
ITEM-12	13,491	103	\$3.24	131.0	2.7	\$43,711	> 90 d (excess)
ITEM-13	4,919	387	\$15.44	12.7	28.3	\$75,949	< 15 d (risk)
ITEM-14	3,550	257	\$8.87	13.8	26.1	\$31,488	< 15 d (risk)
ITEM-15	15,337	317	\$4.17	48.4	7.4	\$63,955	45–90 d (high)

16. The inventory cost structure

Inventory cost taxonomy (slides 42-53)

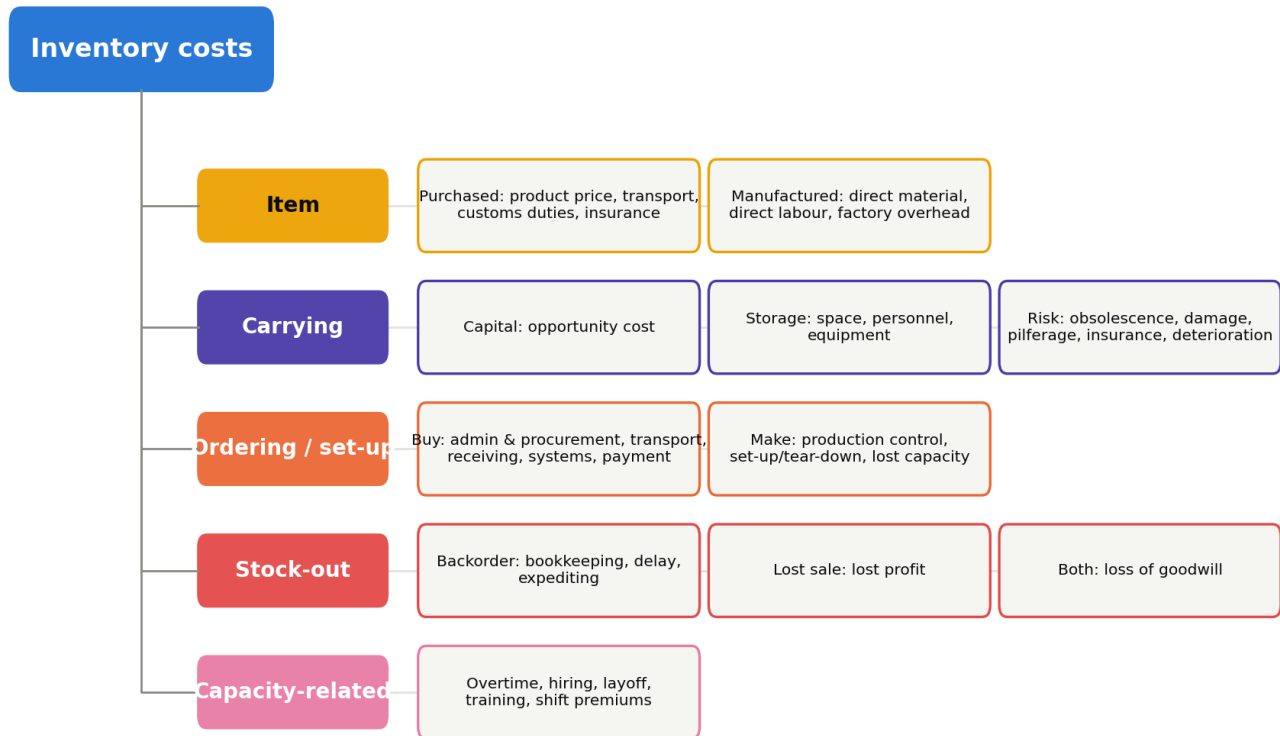


Figure 16.1 — The five cost families and their components.

17. Item (unit) costs of purchased and manufactured items

Purchased: $v = \text{price} + \text{transport} + \text{duties} + \text{insurance}$

Manufactured: $v =$

$\text{DM} + \text{DL} + \text{OH}$

\$735.00

Compressor (purchased)

\$370.00

Washer housing (made)

19.4 %

Avg landed-cost uplift over price

\$46.30

Drum bearing (purchased)

\$258.00

Door assembly (made)

\$483.00

Inverter PCB (purchased)

\$107.00

Wire harness (made)

What one unit of inventory is worth (v)

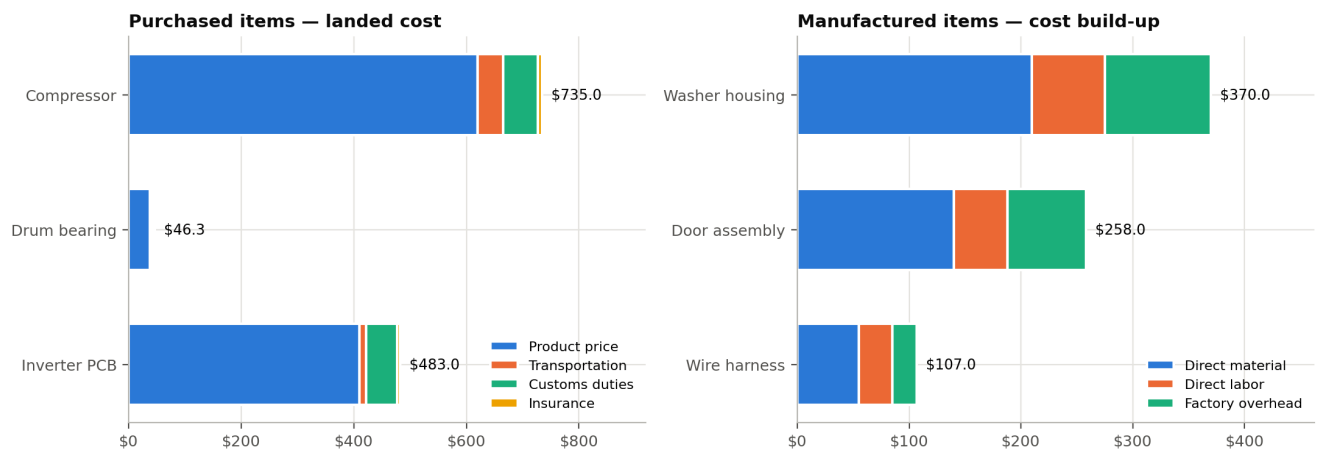


Figure 17.1 — Unit cost components. The unit value v is the basis of the holding cost $h = r \cdot v$.

Item cost build-up (\$ per unit).

item	type	product price	transportation	customs duties	insurance	direct material	direct labor	factory overhead	unit item cost
Compressor	Purchased	620.00	45.00	62.00	8.00	0.00	0.00	0.00	735.00
Drum bearing	Purchased	38.00	3.50	4.20	0.60	0.00	0.00	0.00	46.30
Inverter PCB	Purchased	410.00	12.00	55.00	6.00	0.00	0.00	0.00	483.00
Washer housing	Manufactured	0.00	0.00	0.00	0.00	210.00	65.00	95.00	370.00
Door assembly	Manufactured	0.00	0.00	0.00	0.00	140.00	48.00	70.00	258.00
Wire harness	Manufactured	0.00	0.00	0.00	0.00	55.00	30.00	22.00	107.00

18. Carrying cost: from the annual rate r to the unit holding cost $h = r \cdot v$

$$r = 28\% + 2\% + 6\% + 1\% = 37\% \quad h = 0.37 \times 180 = 66.60 \quad \text{Total} = 5 \times 300 \times 66.60 = 99,900.00$$



Carrying cost — $r = 37\%$ per year (slide 45)

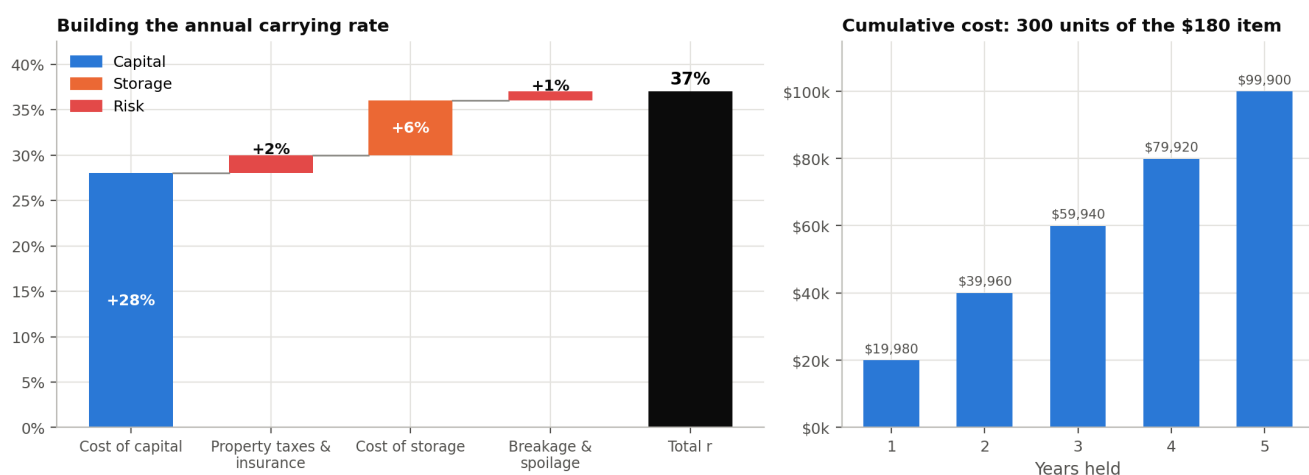


Figure 18.1 — Left: components of r (waterfall). Right: cumulative holding cost of the lecture example.

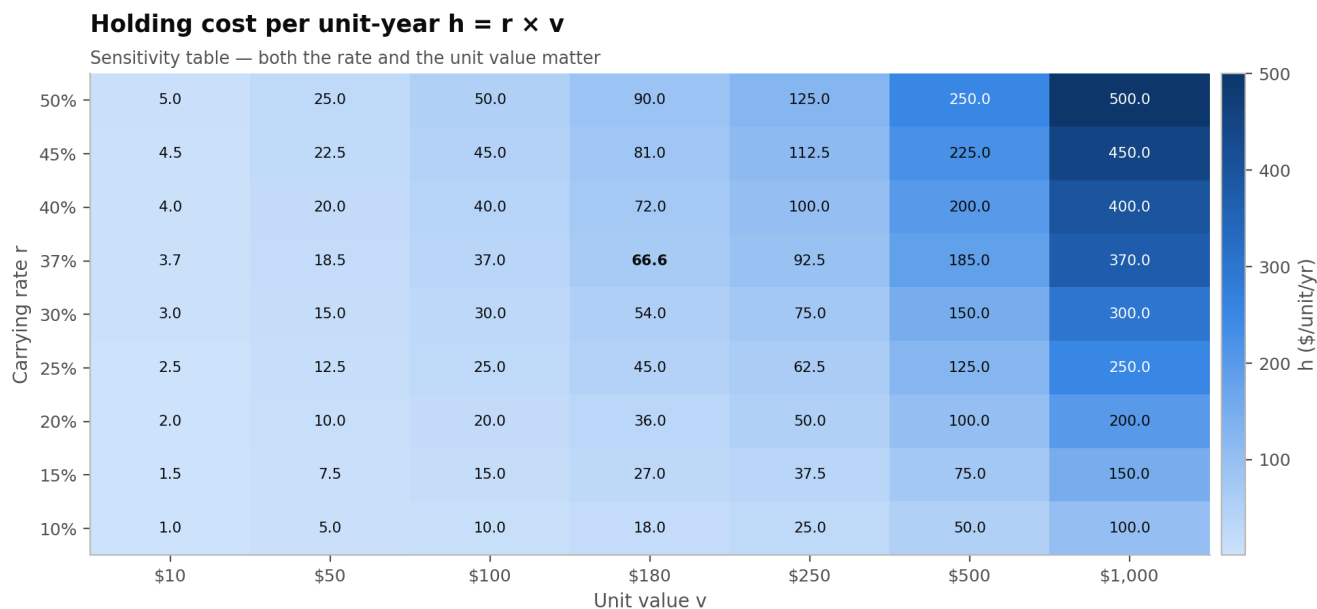


Figure 18.2 — h for combinations of r and v .

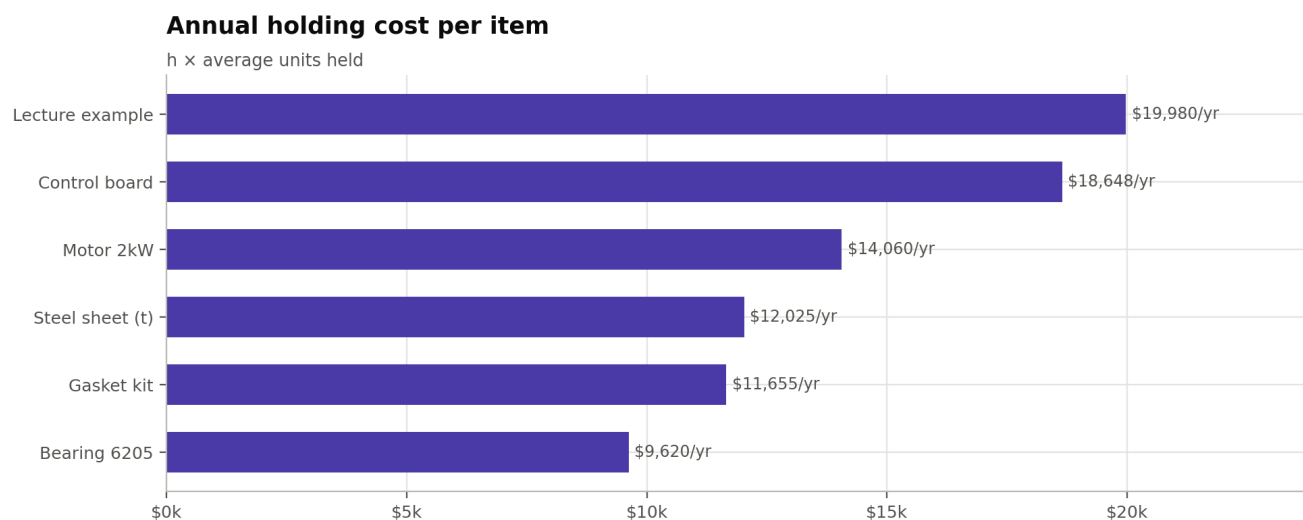


Figure 18.3 — Annual holding cost by item at $r = 37\%$.

Carrying-rate components.

component	group	annual rate pct
Cost of capital	Capital cost	28
Property taxes & insurance	Risk cost	2
Cost of storage	Storage cost	6
Breakage & spoilage	Risk cost	1

Holding cost by item.

item	unit value	avg units held	years held	h per unit year	holding cost per year	total holding cost
Lecture example	\$180.00	300	5	\$66.60	\$19,980	\$99,900
Motor 2kW	\$950.00	40	1	\$351.50	\$14,060	\$14,060
Control board	\$420.00	120	1	\$155.40	\$18,648	\$18,648
Steel sheet (t)	\$1,300.00	25	1	\$481.00	\$12,025	\$12,025
Gasket kit	\$35.00	900	1	\$12.95	\$11,655	\$11,655
Bearing 6205	\$6.50	4,000	1	\$2.40	\$9,620	\$9,620

19. Ordering (buy) and set-up (make) costs

EGP 2,630

Ordering cost per purchase order
(A_{buy})

46%

Lost-capacity share of set-up

EGP 4,150

Set-up cost per production run
(A_{make})

Inbound freight (fixed part)

Largest buy element

Ordering costs are fixed per order — they grow with order frequency

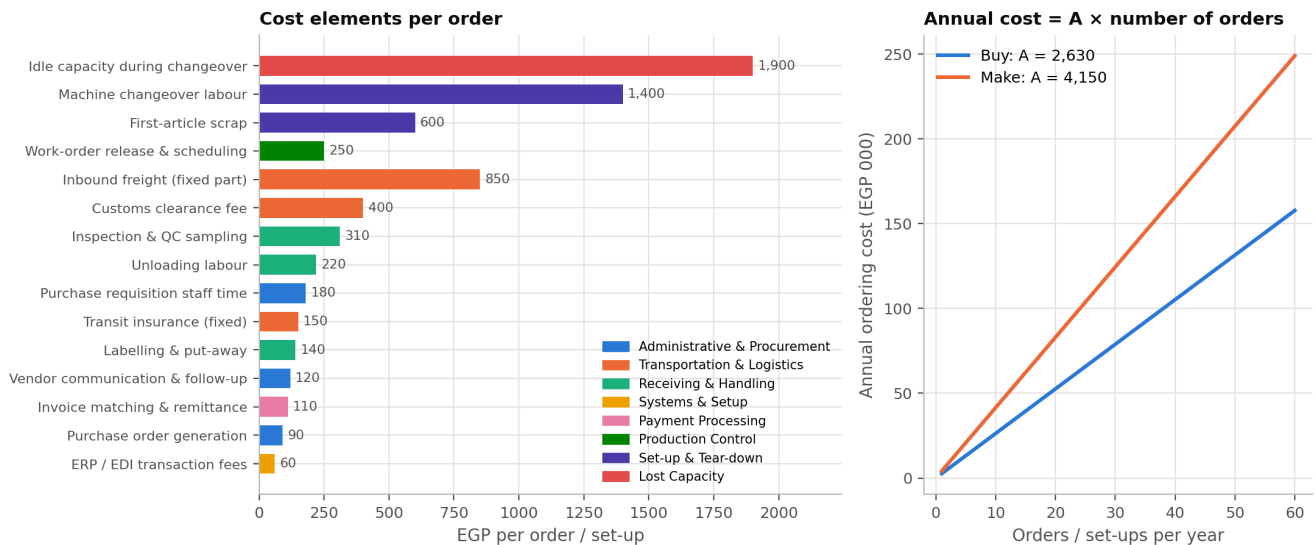


Figure 19.1 — Ordering / set-up cost elements and annual ordering cost vs frequency.

Ordering / set-up cost elements.

mod e	category	element	cost per order egp
Buy	Administrative & Procurement	Purchase requisition staff time	180
Buy	Administrative & Procurement	Purchase order generation	90
Buy	Administrative & Procurement	Vendor communication & follow-up	120
Buy	Transportation & Logistics	Inbound freight (fixed part)	850
Buy	Transportation & Logistics	Customs clearance fee	400
Buy	Transportation & Logistics	Transit insurance (fixed)	150
Buy	Receiving & Handling	Unloading labour	220
Buy	Receiving & Handling	Inspection & QC sampling	310
Buy	Receiving & Handling	Labelling & put-away	140
Buy	Systems & Setup	ERP / EDI transaction fees	60
Buy	Payment Processing	Invoice matching & remittance	110
Make	Production Control	Work-order release & scheduling	250
Make	Set-up & Tear-down	Machine changeover labour	1,400
Make	Set-up & Tear-down	First-article scrap	600
Make	Lost Capacity	Idle capacity during changeover	1,900

Annual ordering cost for selected frequencies.

orders per year	annual cost buy EGP	annual cost make EGP
4	10,520	16,600
12	31,560	49,800
26	68,380	107,900
52	136,760	215,800

20. The holding vs ordering cost trade-off (preview of lot sizing)

Item **Inverter PCB** — $D = 4,800/\text{yr}$, $v = 410$, $A = 2,630$ per order, $r = 37\% \rightarrow h = 151.70$ per unit-year.

$$\text{TRC}(Q) = A \cdot D/Q + h \cdot Q/2$$

408 units

Cost-minimising Q (numerical)

11.8

Orders per year

61,888/yr

Minimum total relevant cost

30,941 vs 30,947

Ordering = holding at optimum?

262–635

Cost within +10 % for Q in

Answering 'how large should the order be?'

Minimum total cost where ordering and holding costs balance — and the curve is flat near the optimum

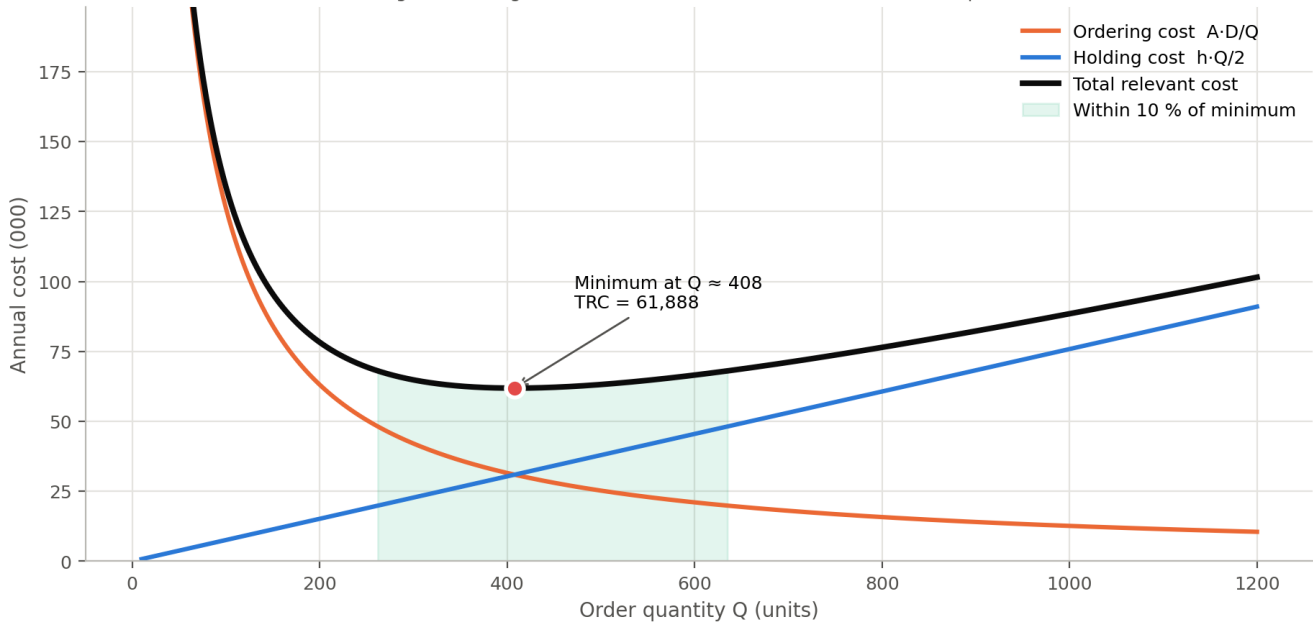


Figure 20.1 — Annual ordering, holding and total relevant cost vs order quantity.

Cost of alternative order quantities.

Q	orders per year	avg cycle stock	ordering cost	holding cost	total cost	pct above min
50	96.0	25	252,480	3,792	256,272	+314.1%
100	48.0	50	126,240	7,585	133,825	+116.2%
200	24.0	100	63,120	15,170	78,290	+26.5%
408	11.8	204	30,941	30,947	61,888	+0.0%
400	12.0	200	31,560	30,340	61,900	+0.0%
800	6.0	400	15,780	60,680	76,460	+23.5%

21. Stock-out costs: backorders vs lost sales

40 (25 backorders, 15 lost sales)

Stock-out events

\$263.70

Avg cost per short unit — lost sale

\$341,604

Total annual stock-out cost

26%

Goodwill share of total

\$60.35

Avg cost per short unit — backorder

15

Expedited events

Where stock-out money goes

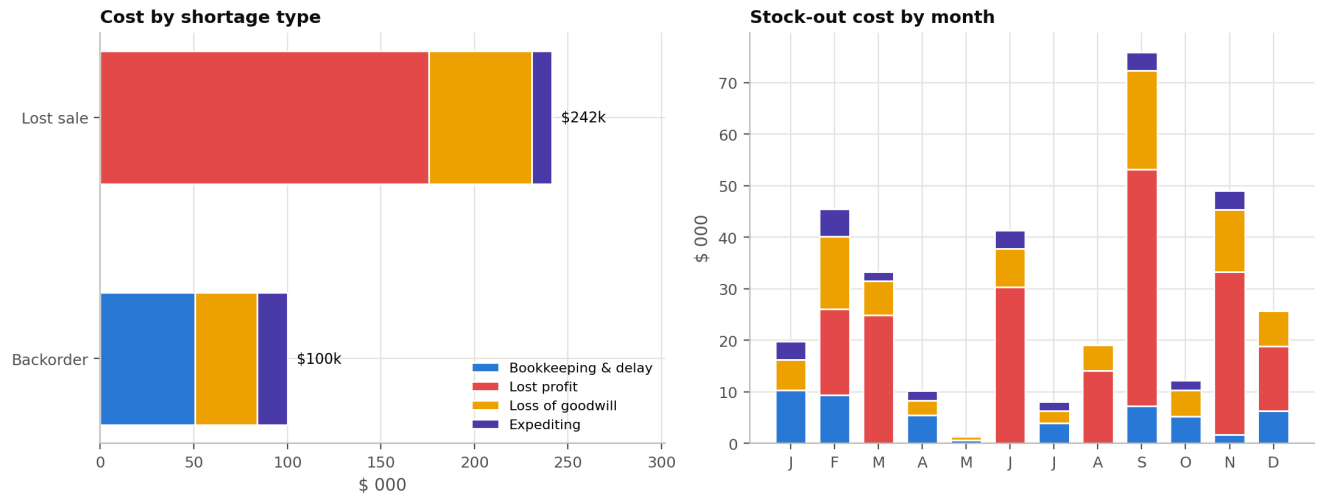


Figure 21.1 — Stock-out cost components by type and by month.

Event cost vs size of shortage

Lost sales cost more per unit (lost margin + larger goodwill loss); large markers = expedited

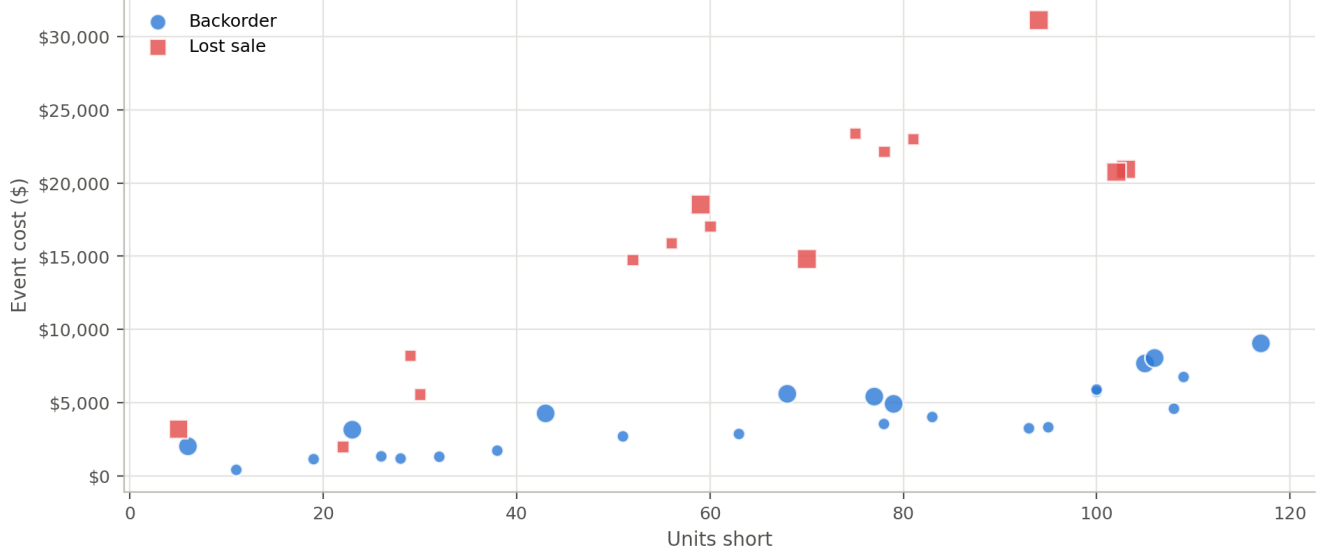


Figure 21.2 — Each point is a stock-out event.

Stock-out cost by item.

item	events	short units	total cost	cost per unit short
Inverter PCB	16	860	\$152,413	\$177.22
Compressor	9	629	\$83,926	\$133.43
Door assembly	9	693	\$82,204	\$118.62

item	events	short units	total cost	cost per unit short
Drum bearing	6	392	\$23,062	\$58.83

Fifteen most expensive stock-out events.

event id	month	item	type	short units	delay days	expedited	bookkeeping delay cost	lost profit	goodwill loss	expediting cost	total cost
25	6	Compressor	lost_sale	94	0	Yes	\$0	\$23,688	\$5,640	\$1,800	\$31,128
28	9	Compressor	lost_sale	75	0	No	\$0	\$18,900	\$4,500	\$0	\$23,400
24	9	Inverter PCB	lost_sale	81	0	No	\$0	\$18,144	\$4,860	\$0	\$23,004
7	11	Inverter PCB	lost_sale	78	0	No	\$0	\$17,472	\$4,680	\$0	\$22,152
2	11	Door assembly	lost_sale	103	0	Yes	\$0	\$12,978	\$6,180	\$1,800	\$20,958
21	2	Door assembly	lost_sale	102	0	Yes	\$0	\$12,852	\$6,120	\$1,800	\$20,772
19	3	Inverter PCB	lost_sale	59	0	Yes	\$0	\$13,216	\$3,540	\$1,800	\$18,556
37	8	Inverter PCB	lost_sale	60	0	No	\$0	\$13,440	\$3,600	\$0	\$17,040
4	12	Inverter PCB	lost_sale	56	0	No	\$0	\$12,544	\$3,360	\$0	\$15,904
26	9	Door assembly	lost_sale	70	0	Yes	\$0	\$8,820	\$4,200	\$1,800	\$14,820
17	3	Inverter PCB	lost_sale	52	0	No	\$0	\$11,648	\$3,120	\$0	\$14,768
33	2	Inverter PCB	backorder	117	20	Yes	\$4,914	\$0	\$2,340	\$1,800	\$9,054
18	6	Inverter PCB	lost_sale	29	0	No	\$0	\$6,496	\$1,740	\$0	\$8,236
38	1	Door assembly	backorder	106	18	Yes	\$4,134	\$0	\$2,120	\$1,800	\$8,054
31	7	Drum bearing	backorder	105	16	Yes	\$3,780	\$0	\$2,100	\$1,800	\$7,680

22. Capacity-related costs: anticipation stock vs changing capacity

Every strategy meets all demand. Regular production cost is included for comparability; subcontracting is costed at its full unit price.



Figure 22.1 — Cost breakdown by strategy.

How each strategy meets the seasonal demand

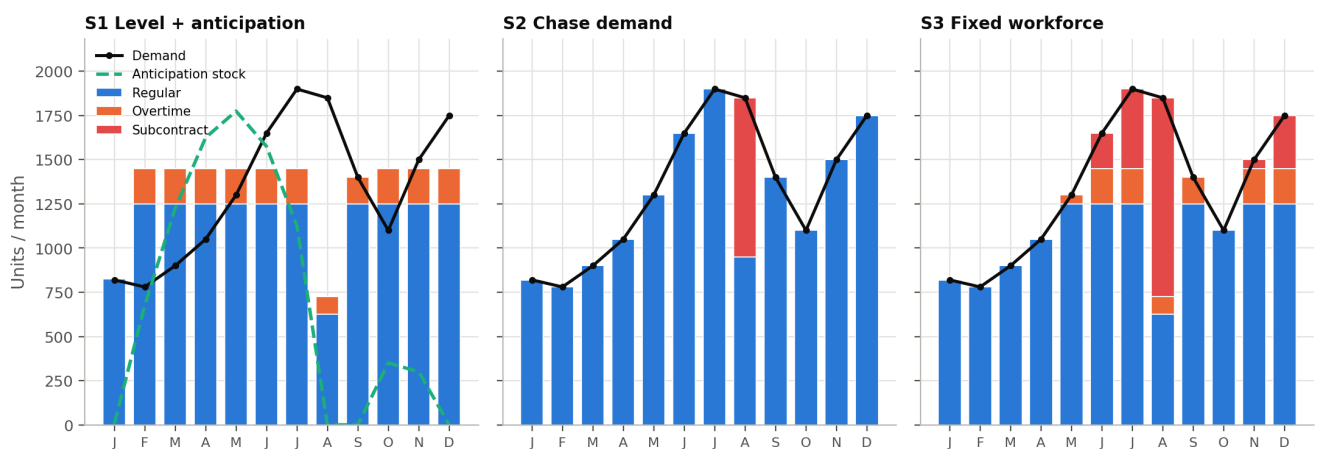


Figure 22.2 — Monthly production by source; dashed = anticipation stock (S1).

Annual cost by strategy and component (\$).

strategy	Regular production	Overtime premium	Holding	Hiring	Training	Layoff	Shift premium	Subcontracting	Total
S1 Level + anticipation	640,000	36,900	19,041	0	0	0	0	0	695,941
S2 Chase demand	604,000	0	0	38,940	14,160	53,100	13,950	67,500	791,650
S3 Fixed workforce	555,000	19,800	0	0	0	0	0	159,375	734,175

23. Other factors: lead time and demand pattern over the life cycle

Replenishment lead time

8 days

Quoted lead time

65%

On-time ($\leq$ quoted)

7.75 days

Mean actual

1,550 units ($d = 200/\text{day}$)

Pipeline stock at mean LT

1.77 days

Std. deviation

10 days

95th-percentile lead time

Replenishment lead time drives pipeline and safety stock

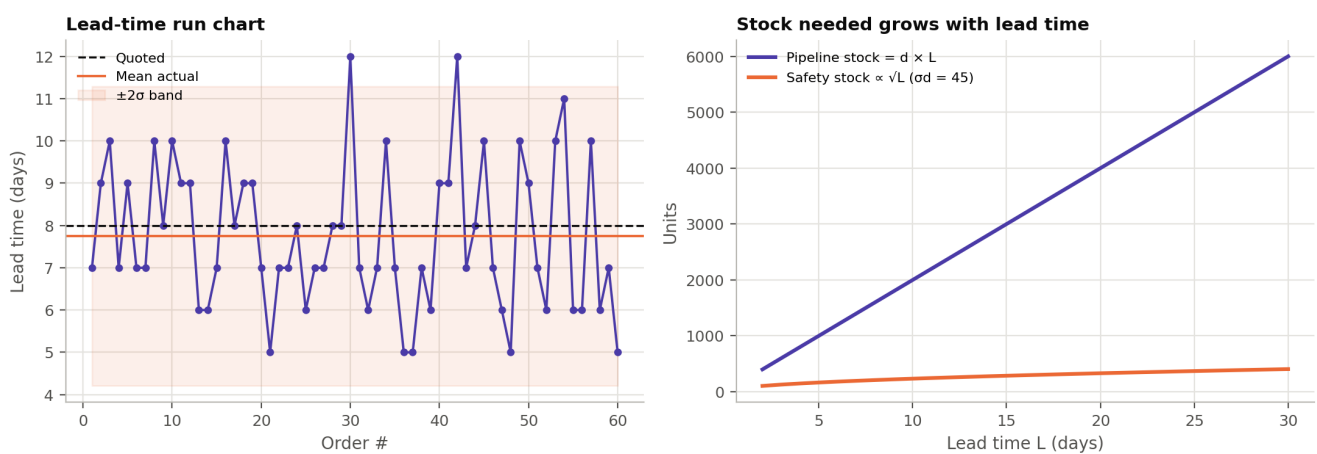


Figure 23.1 — Observed lead times and how pipeline/safety stock scale with the lead time.

Demand pattern and the product life cycle

Different control mechanisms at different life-cycle stages

Illustrative product life cycle

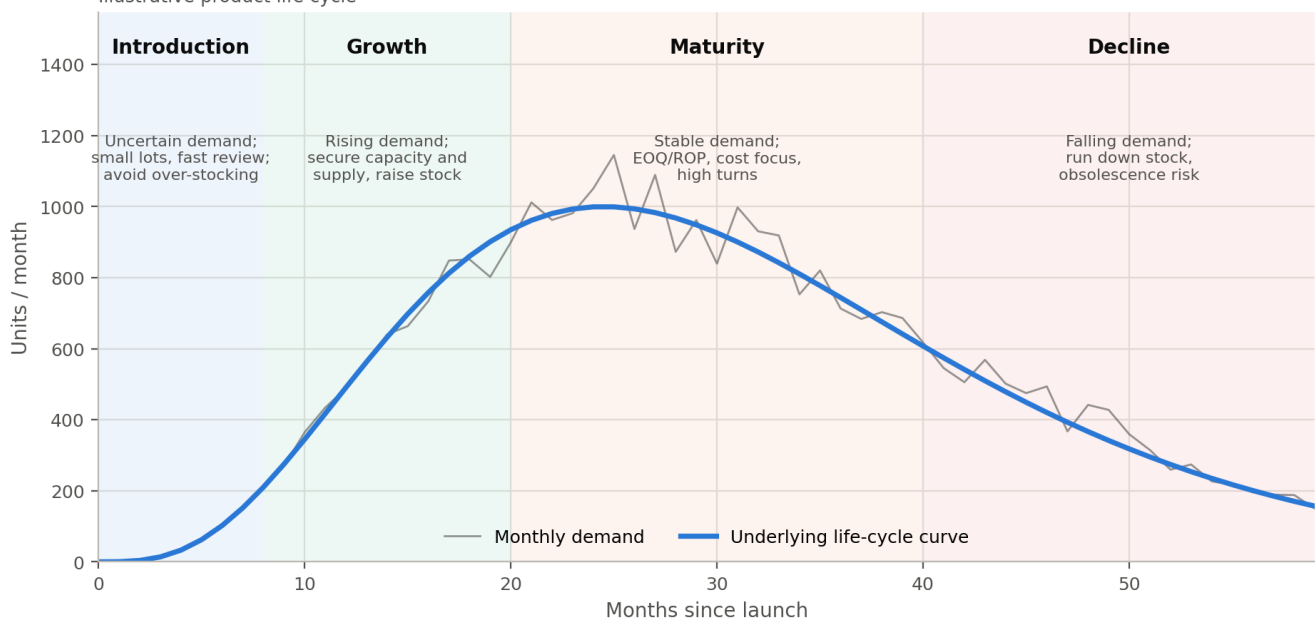


Figure 23.2 — Product life-cycle demand with recommended inventory control emphasis per stage.

Demand statistics by life-cycle stage.

stage	months	avg demand	demand cv	growth per month pct	control emphasis
Introduction	0–8	46	1.14	+45.8	Uncertain demand; small lots, fast review; avoid over-stocking
Growth	8–20	572	0.37	+10.6	Rising demand; secure capacity and supply, raise stock
Maturity	20–40	898	0.15	-2.0	Stable demand; EOQ/ROP, cost focus, high turns
Decline	40–60	366	0.39	-6.7	Falling demand; run down stock, obsolescence risk

Independent vs dependent demand items.

demand type	examples	how determined	typical models
Independent	Finished goods, spare parts, service kits	Forecast from market	EOQ / ROP, periodic review, newsvendor
Dependent	Components, sub-assemblies, raw materials	Calculated from end-item plan × BOM (MRP)	MRP, lot-for-lot, JIT/kanban